Darshan UNIVERSITY

योग: कर्मसु कौशलम्

Career Orientation Aptitude & Reasoning

Subject Material

UG Courses (BTech, BCA, BScIT)

PG Courses (MCA)

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1. Series and Summation

These formulas allow you to calculate sums of large ranges of numbers quickly without manual addition.

- **Sum of first n natural numbers:** $\frac{n(n+1)}{2}$
- **Sum of numbers in a range (e.g., 51 to 100):** Calculate the sum of 1 to 100 and subtract the sum of 1 to 50.
- **Arithmetic Progression (AP) Sum:** $\frac{n}{2}$ (First Term + Last Term)

2. Number System Logic

These techniques solve problems involving unknown integers, remainders, and digit placement.

- **Sum and Difference Rule:** If you know x + y and x - y, solve for x and y by adding or subtracting the equations.
- **LCM and Remainders:** To find the least number that leaves a remainder R when divided by various divisors, find the LCM (divisors) + R.
- **Two-Digit Number Representation:** A two-digit number is 10x + y; interchanging digits results in 10y + x.
- **Constraint Equations:** For "Total Cost" problems (e.g., pineapples and watermelons), set up the equation ax + by = Total and find integer solutions for x and y

3. Basic Algebraic Identities

Use these to simplify complex expressions or find missing values like products or squares.

- $(a+b)^2 = a^2 + b^2 + 2ab$
- $(a-b)^2 = a^2 + b^2 - 2ab$
- $a^2 - b^2 = (a+b)(a-b)$

Exercise

Q.1 (51 + 52 + 53 + ... +100) is equal to **[PI]**

(A) 1325 (B) 5050 (C) 3775 (D) 3325

Q.2 The sum of two number is 25 and their difference is 13. Find their product. **[PI]**

(A) 104 (B) 114 (C) 315 (D) 325

Q.3 A pineapple costs Rs 7 each. A watermelon costs Rs 5 each. X spends Rs 38 on these fruits. The number of pineapples purchased is **[PI]**

(A) 2 (B) 3 (C) 4 (D) 5

Q.4 The least number which when divided by 5, 6 , 7 and 8 leaves a remainder 3, but when divided by 9 leaves no remainder, is **[PI]**

(A) 1677 (B) 1683 (C) 2523 (D) 3363

Q.5 If 144=14.4, then the value of x is:0.144x. **[PI]**

(A) 0.0144 (B) 1.44 (C) 14.4 (D) 144

Q.6 The difference between a two-digit number and the number obtained by interchanging the digit is 36. What is the difference between the sum and the difference of the digits of the number if the ratio between the digits of the number is 1:2? **[PI]**

(A) 4 (B) 8 (C) 16 (D) 20

Q.7 A man has Rs. 480/- in the denominations of one-rupee notes, five-rupee notes and ten-rupee notes. The number of notes of each denomination is equal. What is the total number of notes that he has? **[PI]**

(A) 40 (B) 60 (C) 70 (D) 90

Q.8 If $a - b = 3$, and $a^2 + b^2 = 29$, find the value of ab. **[PI]**

(A) 10 (B) 12 (C) 15 (D) 18

Q.9 A, B and C are partners. They have invested Rs.35000, Rs. 25000 and 10,000 respectively for the same period. If the total profit is Rs. 18000, find the share of A. **[PI]**

(A) 9000 (B) 12000 (C) 8000 (D) 10000

Q.10 X and Y invest Rs. 21000 and Rs. 17500 respectively in a business. At the end of the year, they make a profit of Rs. 26400. What is the share of X in the profit? **[PI]**

(A) 14,400 (B) 26,000 (C) 12,000 (D) 12,500

Q.11 Two numbers have a ratio of 2:3. If their sum is 20, then what is the sum of the squares of two numbers? **[PI]**

(A) 324 (B) 200 (C) 196 (D) None of these

Basic Introduction

- This type of question is totally based on concept of time & distance & speed.
- Mainly problems based on trains have two objects
  - Object 1: Those objects who cross another abject is considered as object 1, e.g. train, bus, person, etc.
  - Object 2: Those objects who are crossed by object 1, e.g. train, platform, over bridge, tunnel, person, pole, tree, wall, etc.

General Rule

1. Concept of speed:
   - When one object is stationary and another is still then only speed of moving object is considered as S_1 = x.
   - When both objects are moving in ***same direction***. In this case relative speed is difference between speed of both objects i.e. ***relative speed = $S_1 - S_2$***.
   - When both objects are moving in ***opposite direction***. In this case relative speed is sum of speed of both objects i.e. ***relative speed = $S_1 + S_2$.***

2. Concept of distance:
   - Distance is always added of both objects with irrespective of direction but in some exception are there when any vertical object is taken then distance of that object is considered as zero. E.g. Length of pole, person and tree is zero.

Formula: $\boldsymbol{S = \frac{L}{T}}$, where S = Speed of train, L = Length of train + Length of object, T = Time taken to cross
Sometimes, we have to convert km/h speed to m/s using below formula

$$1 \text{ km/h} = \frac{5}{18} \text{ m/s}$$

Different types with example

1. When a moving train crosses a standing person

Ex. Find the time taken by a train 150m long, running at 54 km/h in crossing an electrical pole

Sol. $S = 54\ km/h = 54\ X \frac{5}{18} m/s = 15\ m/s$
$L = 150\ m + 0\ m = 150\ m$

$$S = \frac{L}{T}$$

$$\frac{15m}{s} = \frac{150m}{T}$$

$$T = \frac{150ms}{15m}$$

$$T = 10\ s$$

2. When a moving train crosses a moving person in the same direction

Ex. A train 180m long is running at a speed of 65 km/h. In what time will it pass a man who is running at a speed of 5 km/h in the same direction?

Sol. S $= 65km/h - 5km/h = 60km/h$
L $= 180\ m + 0\ m = 180\ m$

$$S = \frac{L}{T}$$

$$\frac{60\ X\ 5\ m}{18\ s} = \frac{180m}{T}$$

$$T = \frac{180\ X\ 18\ ms}{60\ X\ 5\ m}$$

$$T = 10.8\ s$$

3. When a moving train crosses a moving person in the opposite direction

Ex. A train 150m long is running at a speed of 85 km/h. It will cross a man coming from the opposite direction at a speed of 5 km/h.

Sol. S $= 85km/h + 5km/h = 90km/h$
L $= 150\ m + 0\ m = 150\ m$

$$S = \frac{L}{T}$$

$$\frac{90\ X\ 5\ m}{18\ s} = \frac{150m}{T}$$

$$T = \frac{150\ X\ 18\ ms}{90\ X\ 5\ m}$$

$$T = 6\ s$$

4. When a moving train crosses a platform

Ex. A train 140m long is running at a speed of 60km/h. In how much time will it pass a platform 260m long?

Sol. S $= 60km/h$
L $= 140\ m + 260\ m = 400\ m$

$$S = \frac{L}{T}$$

$$\frac{60\ X\ 5\ m}{18\ s} = \frac{400m}{T}$$

$$T = \frac{400\ X\ 18\ ms}{60\ X\ 5\ m}$$

$$T = 24\ s$$

5. When a moving train crosses another stationary train

Ex. A train 180m long takes 20 sec in crossing a stationary train 220m long. The speed of the train is?

Sol. T $= 20\ s$
L $= 180\ m + 220\ m = 400$

$$S = \frac{L}{T}$$

$$\frac{S\ X\ 5\ m}{18\ s} = \frac{400m}{20\ s}$$

$$S = \frac{400\ X\ 18\ ms}{20\ X\ 5\ m}$$

$$S = 72\ km/h$$

6. When a moving train crosses another moving train along same direction

Ex. Two trains travelling in the same direction at 54 km/h and 36 km/h completely pass one another in 1 minute. If the length of the first train is 125 meters, the length of the second train is?

Sol. S $= 54\ km/h\ - 36\ km/h\ = 18\ km/h$
T $= 1\ m = 60\ s$

$$S = \frac{L}{T}$$

$$\frac{18\ X\ 5\ m}{18\ s} = \frac{L}{60\ s}$$

$$L = \frac{18\ X\ 5\ X\ 60\ ms}{18\ s}$$

$$L = 300\ m$$

L $= 300\ m = 125\ m\ + x, \quad x = 300m - 125m = 175m$

7. When a moving train crosses another moving train coming from opposite direction

Ex. Two trains are running in opposite direction with the same speed. If the length of each train is 120 meters and they cross each other in 12 seconds, the speed of each train is?

Sol. T $= 12\ s$
L $= 120\ m + 120\ m = 240\ m$

$$S = \frac{L}{T}$$

$$\frac{S\ X\ 5\ m}{18\ s} = \frac{240m}{12\ s}$$

$$S = \frac{240\ X\ 18\ ms}{12\ X\ 5\ m}$$

$S = 72\ km/h$, hence speed of each train is $36\ km/h$

8. When a moving train crosses a person sitting in another train (both direction)

Ex. A goods train 150 m long moving with speed 54 km/h. A man is sitting in a passenger train which is moving with 18 km/h in same direction. In how many seconds the goods train crosses the man sitting in a passenger train?

Sol. S = $54\ km/h\ - 18\ km/h\ = 36\ km/h$
L = $150\ m + 0\ m = 150\ m$

$$S = \frac{L}{T}$$

$$\frac{36\ X\ 5\ m}{18\ s} = \frac{150m}{T}$$

$$T = \frac{150\ X\ 18\ ms}{36\ X\ 5\ m}$$

$$T = 15\ s$$

9. When a moving train crosses two different objects

Ex. A train running at a speed of 30 km/h takes 18 sec to pass a railway platform and it takes 15 sec to pass a man who is running at a speed of 6 km/h in the same direction. Find the length of platform?

Sol. Here, we have two different cases.
First case is of train and man
S = $30\ km/h\ - 6\ km/h\ = 24\ km/h$
T = $15\ s$

$$S = \frac{L}{T}$$

$$\frac{24\ X\ 5\ m}{18\ s} = \frac{L}{15\ s}$$

$$L = \frac{24\ X\ 15\ X\ 5\ ms}{18\ s}$$

$$L = 100\ m$$

Second case is of train and platform
S = $30\ km/h$
T = $18\ s$, L = $100\ m + length\ of\ platform = 100 + x$

$$S = \frac{L}{T}$$

$$\frac{30\ X\ 5\ m}{18\ s} = \frac{100 + x}{18\ s}$$

$$100 + x = \frac{30\ X\ 18\ X\ 5\ ms}{18\ s}$$

$$x = 150\ m - 100\ m = 50\ m$$

10. When two trains start from two points X & Y towards each other at the same time and after crossing they take p & q second to reach Y & X respectively then ratio of speed = $\sqrt{q} : \sqrt{p}$

Ex. Two trains X and Y start from stations A and B towards B and A respectively. After passing each other, they take 4 hours 48 minutes and 3 hours 20 minutes to reach B and A respectively. If train X is moving at 40 km/h, the speed of train Y is?

Sol. Time taken by X to reach B = 4 hours 48 minutes = 4 x 60 + 48 = 288 minutes
Time taken by Y to reach A = 3 hours 20 minutes = 3 x 60 + 20 = 200 minutes

$$\frac{X's\ speed}{Y's\ speed} = \sqrt{\frac{Time\ taken\ by\ Y\ to\ reach\ A}{Time\ taken\ by\ X\ to\ reach\ B}}$$

$$\frac{40}{Y's\ speed} = \sqrt{\frac{200}{288}} = \sqrt{\frac{100}{144}} = \frac{10}{12} = \frac{5}{6}$$

$$Y's\ speed = \frac{40\ X\ 6}{5} = 48\ km/h$$

Exercise

(Q.1) A train running at 40 km/h crosses a pole in 36 sec. Find the length of train.

(A) 300 m (B) 200 m (C) 400 m (D) 500 m

(Q.2) A train running at the speed of 20 m/s takes 10 sec to pass a man who is running at the speed of 5 m/s in the same direction. Find the length of the train.

(A) 150 m (B) 200 m (C) 250 m (D) 300 m

(Q.3) Two trains are moving in opposite direction at speeds of 80 km/h and 70 km/h and their length is 1.2 km and 0.8 km respectively. Find the time taken by the slower train to cross each other.

(A) 48 s (B) 50 s (C) 42 s (D) 44 s

(Q.4) Two trains are running 40 km/h and 20 km/h respectively in the same direction. The faster train completely passes a man sitting in the slower train in 18 sec. Find the length of faster train.

(A) 105 m (B) 100 m (C) 90 m (D) 110 m

(Q.5) A train 200 m long takes 5 sec to pass a standing man. Find the time taken by train in crossing a railway platform of 240 m long.

(A) 18 s (B) 13 s (C) 11 s (D) 26 s

(Q.6) A man is standing on railway bridge which is 100 m long. He finds that train crossed the bridge in 5 sec but himself in 3 sec. Find the length of the train.

(A) 100 m (B) 150 m (C) 200 m (D) 250 m

(Q.7) A train moves past a telegraph pole and a bridge 264 m long in 8 sec and 20 sec respectively. What is the speed of the train?

(A) 15 m/s (B) 22 m/s (C) 25 m/s (D) 18 m/s

(Q.8) A train overtakes two persons who are walking in the same direction in which the train is going, at the speed of 2 km/h and 4 km/h and passes them completely in 9 sec and 10 sec respectively. Find the length of the train.

(A) 200 m (B) 50 m (C) 150 m (D) 250 m

(Q.9) Two trains each 200 m long move towards each other on parallel track with speed of 22 km/h and 28 km/h respectively. What is the time taken when they meet each other?

(A) 20 s (B) 24.8 s (C) 28.8 s (D) 30 s

(Q.10) A train 240 m long passes a pole in 24 seconds. How long will it take to pass a platform 650 m long?

(A) 65 s (B) 89 s (C) 100 s (D) 150 s

(Q.11) Two trains each 200 m long move towards each other on parallel track with speed of 22 km/h and 28 km/h respectively. What is the time taken when they meet each other? **[PI]**

(A) 20 s (B) 24.8 s (C) 28.8 s (D) 30 s

(Q.12) How many seconds will a 500-meter-long train take to cross a man walking with a speed of 3 km/hr in the direction of the moving train if the speed of the train is 63 km/hr? **[PI]**

(A) 25 (B) 30 (C) 40 (D) 45

(Q.13) A running man crosses a bridge of length 500 meters in 4 minutes. At what speed he is running? **[PI]**

(A) 7.49 Km/hr (B) 7.29 km/hr (C) 7.79 km/hr (D) 7 km/hr

(Q.14) A train 180m long is running at a speed of 65 km/h. In what time will it pass a man who is running at a speed of 5 km/h in the same direction? **[PI]**

(A) 10.8 seconds (B) 11 Seconds (C) 12 Seconds (D) 10 Seconds

(Q.15) Two trains running in opposite directions cross a man standing on the platform in 27 seconds and 17 seconds respectively and they cross each other in 23 seconds. The ratio of their speed is: **[PI]**

(A) 1:3 (B) 3:2 (C) 2:3 (D) 3:4

(Q.16) A train running at the speed of 60 km/hr crosses a pole in 9 seconds. What is the length of the train in meters? **[PI]**

(A) 120 (B) 180 (C) 324 (D) 150

(Q.17) A train 125 m long passes a man, running at 5 km/hr in the same direction in which the train is going, in 10 seconds. The speed (km/hr) of the train is: **[PI]**

(A) 45 (B) 50 (C) 54 (D) 55

(Q.18) The length of the bridge, which a train 130 meters long and travelling at 45 km/hr can cross in 30 seconds, is **[PI]**

(A) 200 m (B) 225 m (C) 245 m (D) 250 m

(Q.19) A train passes a station platform in 36 seconds and a man standing on the platform in 20 seconds. If the speed of the train is 54 km/he, what is the length of the platform? **[PI]**

(A) 120m (B) 240 m (C) 300 m (D) 360 m

(Q.20) A person crosses a 600 m long street in 5 minutes. What is his speed in km per hour? **[PI]**

(A) 3.6 (B) 7.2 (C) 8.4 (D) 10

(Q.21) Two trains are running in opposite directions with the same speed. If the length of each train is 120 meters and they cross each other in 12 seconds, then the speed of each train (in km/hr) is? **[PI]**

(A) 32 (B) 36 (C) 18 (D) 10

(Q.22) It takes eight hours for a 600 km journey, if 120 km is done by train and the rest bycar. It takes 20 minutes more, if 200 km is done by train and the rest by car. Theratio of the speed of the train to that of the car is? **[PI]**

(A) 2:3 (B) 3:4 (C) 4:3 (D) 3:2

General Rule

1. ***Average is equal distribution of the overall value among all the things or persons present there. So, the formula for finding the average is as follows:***
   - Average = Sum of observations / Number of observations

Ex. There are two sections A and B of a class, consisting of 10 and 20 students respectively. If the average weight of section A is 60 kg and that of section B is 45 kg. Find the average weight of the whole class.

Sol. Average = $\frac{60\ X\ 10+45\ X\ 20}{10+20} = \frac{600+900}{30} = 50$

Ex. A batsman makes a score of 97 runs in the 21st inning and thus increases his average by 2. Find his average after 21st inning.

Sol. Before 21st inning, a batsman has played 20 innings and at that time, his average was 2 runs decreased as of now.
- So, he has to score 2 x 20 = 40 runs extra in his 21st inning.
- Hence, after 21st inning, his average becomes 97 – 40 = 57 runs.

Ex. A student was asked to find the arithmetic mean of the numbers 3, 11, 7, 9, 15, 13, 8, 19, 17, 21, 14 and x. He found the mean to be 12. What should be the number in place of x?

Sol. 12 = (3+11+7+9+15+13+8+19+17+21+14+x)/12
144 = 137 + x
x = 7

2. ***Average of consecutive terms:***
   - Average = $\frac{1^{st}\ term+last\ term}{2}$

Ex. Find the average of all the numbers from 20 to 90 which is divisible by 3.

Sol. First term is 21 and last term is 90.

Average = $\frac{21+90}{2} = 55.5$

3. ***Concept of addition or removal of items and change in average:***
   - The excluded or included item = Average ± extra (+ for excluded & - for included)

Ex. Average of 5 no. is 21. If one number is excluded then average becomes 20. What is the excluded no?

Sol. Average is 21. So, if the excluded number is 21, then there will be no change in average.
- But the new average is 20, average decreased by 21 - 20 = 1.
- Total decreased = 1 x 4 = 4, so excluded number = 21+ 4 = 25.

Exercise

Q.1 Out of four members, the average of first 3 numbers is thrice the fourth number. If average of all four members is 5, what is fourth number?

(A) 4.5 (B) 4 (C) 5 (D) 2

Q.2 Average age of students in section A of 55 students is 12 and the average age of students in section B of 45 students is 14 years. Find the average age of students in both sections taken together?

(A) 13 (B) 12.5 (C) 13.2 (D) 12.9

Q.3 The average of six consecutive odd no is 48. Then find out largest and smallest no.

(A) 53, 43 (B) 53, 41 (C) 55, 45 (D) 57, 47

Q.4 Find the average of numbers lies between 100 and 300 as well as divisible by 7.

(A) 200 (B) 199.5 (C) 205 (D) 199

Q.5 The average weight of 29 students is 40 kg. If the weight of teacher be included the average weight increased by 400 gms. The weight of the teacher in kg is?

(A) 52 (B) 55 (C) 50 (D) 49

Q.6 Average marks of 35 students is 70. If marks of one student is excluded then average increased by 2 marks. What are the marks of excluded student?

(A) 2 (B) 4 (C) 6 (D) 8

Q.7 The average weight of 10 persons is increased by 2.5 kg when two of them having weight 60 and 75 kg replaced by two new persons. Then what is the average weight of new persons?

(A) 67.5 (B) 70 (C) 80 (D) 72.5

Q.8 The average of 50 numbers is 38. It was found that one number is misread as 89 instead of 39. The correct mean is?

(A) 37 (B) 36 (C) 37.2 (D) 36.6

Q.9 The average of 10 numbers is 43. Average of first four is 40 and last five is 47. Then 5th number is?

(A) 38 (B) 35 (C) 32 (D) 36

Q.10 In the first 10 overs of a cricket game, the run rate was 6.8. What should be the run rate in the remaining 40 overs to reach the target of 350 runs?

(A) 6.05 (B) 5.05 (C) 7.05 (D) 8.05

Q.11 The captain of the cricket team of 11 members is 26 years old and the wicket keeper is 3 years older than captain. If the ages of these two are excluded, the average of the remaining players is one year less than the average age of the whole team. What is the average age of the team? **[PI]**

(A) 23 Years (B) 24 Years (C) 25 Years (D) None of these

Q.12 In the first 10 overs of a cricket game, the run rate was only 3.2. What should be the run rate in the remaining 40 overs to reach the target of 282 runs? **[PI]**

(A) 6.25 (B) 5.5 (C) 7.4 (D) 5

Q.13 What is the average of the first five multiples of 12? **[PI]**

(A) 36 (B) 48 (C) 24 (D) 60

Q.14 Out of four numbers, the average of the first 3 numbers is thrice the fourth number. If the average of all four numbers is 5, what is the fourth number? **[PI]**

(A) 2 (B) 4 (C) 5 (D) 6

Q.15 The average age of 5 children born at the intervals of 3 years each is 50 years. What is the age of the youngest child? **[PI]**

(A) 42 Years (B) 44 Years (C) 46 Years (D) 48 Years

Q.16 If the median of the observations 2, 3, 5, 6, x, 8, 9 is 6 the value of x cannot be equal to? **[PI]**

(A) 8 (B) 1 (C) 10 (D) 15

Q.17 The average weight of 8 persons increases by 2.5 kg when a new person comes in place of one of them weighing 65 kg. What will be the weight of the new person? **[PI]**

(A) 76 (B) 76.5 (C) 85 (D) 80

Q.18 The average marks in science subject of 20 students are 68. If the marks of two students were misread as 48 and 65 instead of the actual marks i.e. 72 and 61 respectively. Find the correct average? **[PI]**

(A) 69.5 (B) 66 (C) 67 (D) 69

Introduction

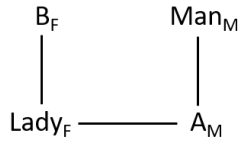
Symbols used for Relation representation

- X_M = indicates X is male
- X_F = indicates X is female
- X ⟷ Y = Husband wife
- X – Y = Sibling
- X = Son or Daughter
  |
  Y

Type-1: Indicating form

Ex. Introducing a man, a lady said that " He is the father of my mother's daughter's only brother". How is the man related to lady?

Sol.



- The man is father of that lady.

Type-2: Puzzle form

Ex. P, Q, R, S, T and U are six members of a family. There are two married couples among them and total members of female in the family is three.

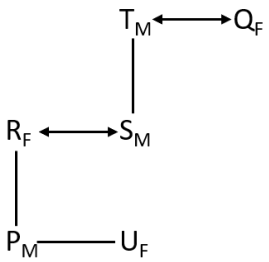
- R is the mother of P and U
- T is the father of S
- P is the grandson of T

Q-1 How Q is related to U?

Q-2 Who is daughter of S?

Q-3 If X is married to U then how X is related to S?

Sol.

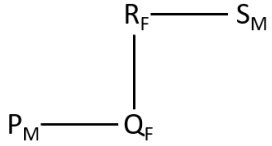


- Answer - 1: Grandmother
- Answer - 2: U
- Answer: Son-in-law

Type-3: Coding form

Ex. 'A+B' means 'A is the father of B'
'A-B' means 'A is wife of B'
'AxB' means 'A is the brother of B'
'A/B' means 'A is the daughter of B'
In which of the following means 'S is father of P'
(1) P/Q-RxS (2) PxQ/R-S (3) P-QxR/S (4) P/Q+RxS

Sol.



- PxQ/R-S is the correct answer.

Exercise

(Q.1) Pointing to the lady in the photograph, Mitali said "Her son's father is the only son-in-law of my mother." How is Mitali related to the lady?

(A) Sister (B) Mother (C) Cousin (D) Self

(Q.2) Pointing to a photographer of Prabhu, Vijay said "The father of his sister is the husband of my wife's mother." How is Vijay related to Prabhu?

(A) Brother (B) Brother-in-law (C) Uncle (D) Father

(Q.3) Pointing towards a person in a photograph, Shivani said "He is the only son-in-law of the father of my brother." How is that person is related to Shivani?

(A) Father-in-law (B) Brother (C) Husband (D) Mother

(Q.4) P is the father of Q, who is the father of A. B is the daughter of R, who is the only daughter-in-law of P. P has one grandson and one granddaughter. Then how is P related to A?

(A) Grandfather (B) Grandmother (C) Father (D) Mother

(Q.5-6) There are six persons A, B, C, D, E and F in a family. C is the sister of F. B is the brother of E's husband. D is the father of A and grandfather of F. There are two fathers, three brothers and a mother in the group.

(Q.5) Which of the following is a group of brothers?

(A) ABF (B) ABD (C) BFC (D) HEW

(Q.6) How is C related to D?

(A) Granddaughter (B) Daughter-in-law (C) Grandson (D) Son-in-law

(Q.7-10) Study the following information carefully and answer the questions given below:
'P$Q' means 'P is the father of Q'
'P@Q' means 'P is the mother of Q'
'P#Q' means 'P is the son of Q'
'P&Q' means 'P is the sister of Q'
'P*Q' means 'P is the wife of Q'

(Q.7) Which of the statement means A is the father-in-law of D?

(A) A*B#C$D@E (B) A$C@E#D$B (C) C@B#A$D*E (D) D$A@B*C#E

(Q.8) Which of the statement is true about the given expression, G*T$Y&I*J?

(A) T is the mother of I (B) G is the husband of T
(C) Y is the sister-in-law of J (D) T is the mother-in-law of J

(Q.9) In the given expression, 'M&N*O#P@R', how is M related to O?

(A) Daughter-in-law (B) Sister-in-law (C) Aunt (D) Mother

(Q.10) How is N related to O, according to the given expression, 'M@N&P*Q$O'?

(A) Father-in-law (B) Father (C) Mother (D) Aunt

(Q.11) There is one married couple in the family. A and B are a married couple, A being the male member. D is the only son of C who is the brother of A. E is the sister of D. B is the daughter-in-law of F whose husband has died. How is E related to C. **[PI]**

(A) Son (B) Grandson (C) Daughter (D) Son-in-law

(Q.12) Introducing a boy, a girl said, "He is the son of the daughter of the father of my uncle." How is the boy related to the girl? **[PI]**

(A) Brother (B) Nephew (C) Uncle (D) Son-in-law

(Q.13) Introducing a man, a lady said that "He is the father of my mother's daughter's only brother". How is the man related to the lady? **[PI]**

(A) Father (B) Son (C) Brother (D) None

(Q.14) A is B's sister. C is B's mother. D is C's father. E is D's mother. Then, how is A related to D? **[PI]**

(A) Grandfather (B) Grandmother (C) Granddaughter (D) Grear Grandmother

(Q.15 – 19) A, B, C, D, E, F and G are the seven members of a family. There are two married couple and two children in the 3rd generation. **[PI]**

G is B's mother.

D is E's mother.

F is C's son

B is E's Aunt.

How is D related to B? **[PI]**

(A) Sister (B) Sister-in-law (C) Mother (D) Aunt

If C is B's husband, how is F related to E? **[PI]**

(A) Brother (B) Sister (C) Cousin (D) Can not be determine

If A is G's son, how is A related to F? **[PI]**

(A) Uncle (B) Father (C) Cousin (D) Brother

Who is E's father? (Use the data, if necessary, from the previous question) **[PI]**

(A) B (B) A (C) C (D) G

How many females are there in the family? (Use the data, if necessary, from the previous question) **[PI]**

(A) 3 (B) 4 (C) 5 (D) Can not be determine

(Q.20-24) Blood Relations for a family consisting of L, M, N, R, P, Q is given below. Answer the following questions based on the relation given. **[PI]**

R is the daughter of L.

Q is the brother of L.

N and L are married couples.

P is the brother of N.

M is the son of N but N is not the mother of M.

Find the maternal uncle of R? **[PI]**

(A) Q (B) M (C) N (D) P

Who is the brother of R? **[PI]**

(A) L (B) M (C) N (D) P

Which person in the following option does not belong to the same generation of Q? [PI]

(A) P (B) N (C) M (D) L

How many female members are there in the family? [PI]

(A) 1 (B) 2 (C) 3 (D) 4

How many male members are there in the family? [PI]

(A) 1 (B) 2 (C) 3 (D) 4

(Q.25) If A + B means A is the brother of B; A - B means A is the sister of B and A x B means A is the father of B. which of the following means that C is the son of M. [PI]

(A) M - N x C + F (B) F - C + N x M (C) N + M - F x C (D) M x N - C + F

(Q.26) A + B means A is the sister of B

A x B means A is the son of B

A% B means B is the daughter of A

Now, consider the following relation: M + N% O+R

How is M related to R? [PI]

(A) Aunt (B) Sister-in-law (C) Sister (D) Mother

(Q.27) There are six individuals. Mittal, Priya, Komal, Kiran, Suresh, Kiran, and Dipen. Mittal's sister is Komal. Priya's husband's brother is Kiran. Suresh's father is Dipen, and Mittal's grandpa is Dipen. The group consists of a mother, three brothers, two fathers, and two brothers. The mother, who is she? [PI]

(A) Priya (B) Komal (C) Suresh (D) Komal

(Q.28) Pointing out to a lady, a girl said, "She is the daughter-in-law of the grandmother of my father's only son." How is the lady related to the girl? [PI]

(A) Aunt (B) Mother (C) Can not be determined (D) Sister-in-law

(Q.29) A is the sister of B. B is the brother of C. C is the son of D. How is D related to A? [PI]

(A) D is the father of A (B) D is the mother of A (C) D is the parent of A (D) None of these

(Q.30) A3P means A is the mother of P A4P means A is the brother of P A9P means A is the husband of P A5P means A is the daughter of P Which of the following means that K is the mother-in-law of M? [PI]

(A) M9N3K4J (B) M9N5K3J (C) K5J9M3N (D) K3J9N4M

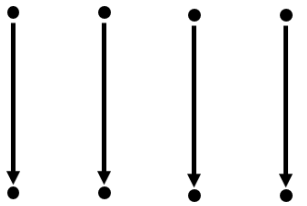
(Q.31) Pointing to a woman, Abhijit said, "Her granddaughter is the only daughter of my brother." How is the woman related to Abhijit? [PI]

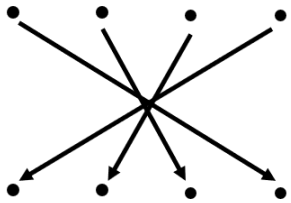
(A) Sister (B) Grandmother (C) Mother-in-law (D) Mother

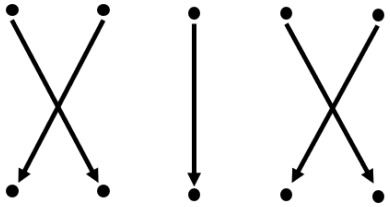
Introduction

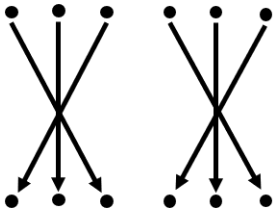
- It is a technique to transform the characters in such a way that no third person can understand it.
- Approach of solving the question of coding:
  a) Observe characters (alphabets/numbers/symbols) given in the code properly.
  b) Find the pattern and sequence it follows, whether it is ascending, descending or pair based.
  c) Solve the asked characters using the same pattern and sequence.

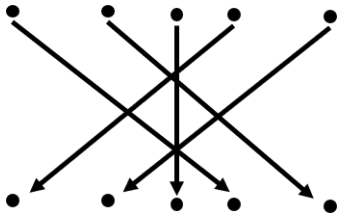
Different types of pattern for coding

(a) 

(b) 

(c) 

(d) 

(e) 

Alphabets with numbers

1	2	3	4	5	6	7	8	9	10	11	12	13
A	B	C	D	E	F	G	H	I	J	K	L	M
26	25	24	23	22	21	20	19	18	17	16	15	14

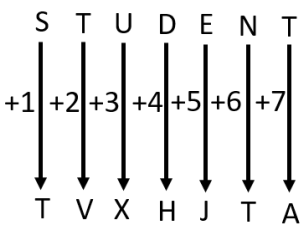
14	15	16	17	18	19	20	21	22	23	24	25	26
N	O	P	Q	R	S	T	U	V	W	X	Y	Z
13	12	11	10	9	8	7	6	5	4	3	2	1

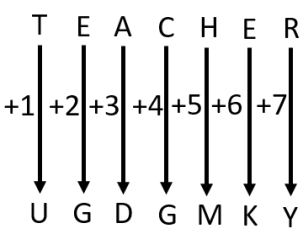
Letter Coding

- In letter coding, letters are coded in letters by using different patterns and sequence.

Ex. In a certain code language STUDENT is written as TVXHJTA, then how is TEACHER written in same code language?

Sol.





Number Coding

- In number coding, letters are coded in numbers by using different patterns and sequence.

Ex. In a certain code language ASKED is written 45211, how will EIGHT be written in that code language?

Sol. A = 1, S = 19, K = 11, E = 5, D = 4

- Convert all digits to single digit, we will get 1, (1+9=10, 1+0=1), (1+1=2), 5, 4
- Therefore, number is 1 1 2 5 4. Write it in reverse order "45211"
- Same pattern is used for EIGHT as shown below
- E = 5, I = 9, G = 7, H = 8, T = 20
- Convert all digits to single digit, we will get 5, 9, 7, 8, (2+0 = 2)
- Therefore, number is 5 9 7 8 2. Write it in reverse order "28795"

Conditional Coding

- In conditional coding, few conditions are given in the question, we have to follow the same to code the characters.

Ex. In each of these questions a group of letters is given followed by four combinations of number/symbol numbered (1), (2), (3) & (4). Letters are to be coded as per the scheme and conditions given below. You have to find out the serial number of the combination, which represents the letter group. Serial number of that combination is your answer.

Letter: E Q B K N P L I T C S F H W A

Symbol: 5 * $ 2 © # 4 9 @ 6 1 8 % 7 3

Conditions:

(1) If the first letter is a consonant and the last letter is a vowel, both will be treated as vowel.
(2) If the first letter is a vowel and the last letter is a consonant, the codes will be interchange.
(3) If both the first and the last letters are consonants, both will be coded as 'd'.
(4) If there are more than two vowels in the group of letters, then all vowels will be coded as '&'.

Q-1 IQCPWF

Sol. According to condition (2), the code is 8*6#79

Q-2 KAWILP

Sol. According to condition (3), the code is d379#d

Q-3 TCKAPE

Sol. According to condition (1), the code is 5623#5

Q-4 IBTNAE

Sol. According to condition (4), the code is &$@©&&

Q-5 IKBQFA

Sol. No condition is satisfied, the code is 92$*83

Exercise

(Q.1) In a certain code language, REASONING is written as HZVILTMRM, then how is DIGINOTES written in same code language?

(A) RTRWMHVGL (B) RTYWNHMGL (C) RTRMWKVGL (D) RLRHMWVGL

(Q.2) In a certain code language ZEBRA is written as WBYAJ, then how is HORSE written in same code language?

(A) MITIW (B) MWITI (C) MJTKP (D) MTIWI

(Q.3) In a certain code language, GGOLD is written as IKUTN, then how is STARS written in same code language?

(A) UXGZS (B) UXFZC (C) UXGZC (D) UWGXC

(Q.4) In a certain code language, RTS is written as 19 and MTH is written as 25, then how is IPL written in same code language?

(A) 13 (B) 12 (C) 23 (D) 17

(Q.5) In a certain code language, PRINTER is written as STJNSCO, then how is DIGITAL written in same code language?

(A) GHISYIK (B) HGISIYK (C) GKHISYI (D) KGHISYI

Q.6-10 In each of these questions a group of letters is given followed by four combinations of number/symbol numbered (1), (2) & (3). Letters are to be coded as per the scheme and conditions given below. You have to find out the serial number of the combination, which represents the letter group. Serial number of that combination is your answer.

Letter: S P W J K I L T Q O D R Y H E A F G U

Symbol: x + 1 = # 4 ? $ @ 5 6 3 8 9 2 o 7 / ©

Conditions:

(1) If the first and last letter place's value are even number according to the English alphabetical order then both are coded by the code of U.
(2) If the first and last letter place's value are odd number according to the English alphabetical order then both are coded by the code of S.
(3) If the first letter is a vowel and last letter is a consonant and there are more than two vowels in the group then both first and last letters are coded by each other codes.

(Q.6) EDAIFH

(A) 96o472 (B) /6o473 (C) 7/6347 (D) 4/6752

(Q.7) QERTYW

(A) x23$84 (B) x23$8x (C) x25$8x (D) x23$1x

(Q.8) RWYHPJ

(A) 4189+© (B) ©179+© (C) ©183+© (D) ©189+©

(Q.9) TQODSP

(A) ©@5xx© (B) ©@57x© (C) ©@56x© (D) ©@36x©

(Q.10) SWJKIL

(A) x1=#4? (B) x13#4? (C) x1=?4? (D) x1+#4?

(Q.11) In a certain code language if the word 'MUSEUM' is coded as 'LSPAPG', then how will the word 'PALACE' be coded in that language? **[PI]**

(A) OYIWXY (B) OYIXYW (C) IYXYWO (D) YXWYOI

(Q.12) In a certain code language, if the number 1 is assigned to all the letters in odd numbered places in the alphabet and the remaining letters are assigned the number 2, then what is the code for the word 'DANCE'? **[PI]**

(A) 21211 (B) 12121 (C) 22111 (D) 21121

(Q.13) In a certain code language, if the word 'DISTANCE' is coded as EDCINSAT, then how will you code 'ACQUIRE' in that language? **[PI]**

(A) EACIQUR (B) EACRIUQ (C) ERCIAQU (D) EARCIQU

(Q.14) In a certain code language, if the value of 28+14=50 and 36+43=63, then what is the value of 44+52=? **[PI]**

(A) 54 (B) 56 (C) 58 (D) 62

(Q.15) These questions are based on a certain code language. Understand the logic in the coding and answer the following questions. **[PI]**

"PROCESSOR" is coded as D4F3C5C1E1S1S1E3C6

Find code for "QUADRANT"

(A) Q1C7A1B2F6 A1B7E4 (B) Q1D6A1B2F3 A1B7E4 (C) Q1C7A1B2F3A 1B7E4 (D) Q1C7A1D2C6A1 B7E5

(Q.16) In a certain code language, 'kewxashumadeko' means 'she is eating apples'; 'kewtepo qua' means 'she sells toys' and 'sullimdeko' means 'I like apples'. Which word in that language means 'she' and 'apples'? **[PI]**

(A) xas&deko (B) xas&kew (C) kew&deko (D) kew&xas

(Q.17 - Q.21) Study the following arrangement carefully and answer the questions given below: 1 to 5

B # b A R 5 8 E % M F 4 J 1 U @ H 2 9 T 1 6 9 * W 3 P # K 7 $ Y

(Q.17) If all the numbers in the above arrangement are dropped, which of the following will be the eleventh from the right end? **[PI]**

(A) U (B) T (C) F (D) H

(Q.18) How many such consonants are there in the above arrangement each of which is immediately preceded by a consonant and also immediately followed by a number? **[PI]**

(A) 1 (B) 2 (C) 3 (D) More than 3

(Q.19) How many such symbols are there in the above arrangement each of which is immediately preceded by a letter and also immediately followed **[PI]**

(A) 1 (B) 2 (C) 3 (D) More than 3

(Q.20) How many such numbers are there in the above arrangement each of which is immediately preceded by a consonant and also immediately followed by a symbol? **[PI]**

(A) 1 (B) 2 (C) 3 (D) More than 3

(Q.21) Which of the following is the twelfth to the left of the twentieth from the left end of the above arrangement? **[PI]**

(A) E (B) W (C) S (D) J

(Q.22) If A is coded as +, B is coded as C is coded as and D is coded as / then valuate 7 A 7 B 2 C 2 7 D 3 C 4? **[PI]**

(A) -58 (B) -56 (C) -48 (D) -36

(Q.22) If A is coded as, B is coded as +, C is coded as and D is coded as/then evaluate 5 A 8 B 6 C 5 D 2 C 3 7 **[PI]**

(A) 43 (B) 44 (C) 42 (D) 52

(Q.23) If "*" means -, "+" means /, "." means * and, "/" means +, then 15 - 2 / 900 + 90 * 100 = ? **[PI]**

(A) -30 (B) -40 (C) -50 (D) -60

(Q.24) If 'waste management techniques' is coded as '25 19 16' in the given language, then how will 'farming fertilizer management' be coded as? **[PI]**

(A) 25, 12, 11 (B) 13, 25, 14 (C) 12, 13, 25 (D) None of These

(Q.25) What will be the code for 'few waste' in the given code language? **[PI]**

(A) 16, 17 (B) 17, 13 (C) 14, 19 (D) None of These

(Q.26) In a certain code "COMPUTER" is written as "RFUVQNPC". How will be "SCIENCE" written in that code? **[PI]**

(A) HVXRMWV (B) HWXRMWU (C) HXDQMWV (D) HVXRMUV

(Q.27) If "PROGRAMMING" is coded as "QSPHSBNNJOH", what is "ALGORITHM" coded as? **[PI]**

(A) BMHPSJUIN (B) BMHPSJUIO (C) BMIPSUJIN (D) CMIPSUJIN

(Q.28) MONKEY = 74, TIGER = 54, LION = ? **[PI]**

(A) 44 (B) 45 (C) 46 (D) 50

Introduction

- In most of the problems on time and work, one of the following basic parameters is to be calculated:
  (a) Time: Time needed by one or more than one person to complete a job or time for which a person actually worked on the assigned job.
  (b) Work: The amount of total work assigned or the part of total work actually done.

Type-1: When more than one person working on the same piece of work.

Ex. A can do a piece of work in 6 days and B in 9 days. How many days will both take together to complete the work?

Sol.

Person	No. of days to complete the work alone	LCM (Total work, mostly we can assume water tank (in liters)	Work done by person in one day (Liters/day)
A	6	18	3 (18/6)
B	9		2 (18/9)

- If both persons work together, then work done in one day is 5 liters (3+2).
- Therefore, together they complete the whole work in 18/5 days.

Ex. If man A complete a piece of work in 12 days and the same work is completed by B and C in 15 days and 20 days respectively. In how many days the work is completed if they work together?

Sol.

Person	No. of days to complete the work alone	LCM (Total work, mostly we can assume water tank (in liters)	Work done by person in one day (Liters/day)
A	12	60	5 (60/12)
B	15		4 (60/15)
C	20		3 (60/20)

- If all three persons work together, then work done in one day is 12 liters (5+4+3).
- Therefore, together they complete the whole work in 60/12 = 5 days .

Ex. A and B together can do a piece of work in 5 days and A alone can do it in 8 days. B alone can do the same piece of work in?

Sol.

Person	No. of days to complete the work alone	LCM (Total work, mostly we can assume water tank (in liters)	Work done by person in one day (Liters/day)
A+B	5	40	8 (40/5)
A	8		5 (40/8)

- If B alone do this work, then work done in one day is 3 liters (8 - 5).
- Therefore, B alone completes the whole work in 40/3 days.

Ex. A and B can complete a piece of work in 8 days, B and C can do it in 12 days, C and A can do it in 8 days. A, B and C together can complete it in?

Sol.

Person	No. of days to complete the work alone	LCM (Total work, mostly we can assume water tank (in liters)	Work done by person in one day (Liters/day)
A+B	8	24	3 (24/8)
B+C	12		2 (24/12)
C+A	8		3 (24/8)

- So, 2(A+B+C) work per one day is 8 liters (3+2+3) and together their wok per day is 4 liters.
- Therefore, together they complete the whole work in 24/4 = 6days.

Ex. A can complete a piece of work in 20 days & B can complete the same work in 25 days. If they start work together but after 5 days A left the work. In how many days the total work would be finished?

Sol.

Person	No. of days to complete the work alone	LCM (Total work, mostly we can assume water tank (in liters)	Work done by person in one day (Liters/day)
A	20	100	5 (100/20)
B	25		4 (100/25)

- If both persons work together, then work done in one day is 9 liters (5+4).
- For the first 5 days, they work together and completed work is 9x5 = 45 liters.
- Now, for the remaining work (i.e. 55 liters) only B works alone.
- Therefore, B alone works for 55/4 days to complete the remaining work.
- Total days to complete the work is 5 + 55/4 = 75/4 days.

Ex. A is twice as good a work man as B & together they finish a piece of work in 18 days. In how many days will A alone finish the work?

Sol. Here, the work efficiencies of A & B are given, A : B = 2 : 1

- Time taken by A and B together = 18 days
- Work done by A and B = 18 x 3 (total of efficiencies) = 54 liters
- A will do work alone = 54/2 = 27 days

Type-2: If M_1 can do W_1 work in D_1 days working H_1 hr/day for Rs. R_1 and M_2 can do W_2 work in D_2 days working H_2 hr/day for Rs. R_2, then

$$\frac{M_1 D_1 H_1}{W_1 R_1} = \frac{M_2 D_2 H_2}{W_2 R_2}$$

Ex. If 10 men complete half work in 12 days when they work 8 hours per day. In how many days 18 men complete the full work when they work 6 hours per day?

Sol. $M_1 = 10$, $W_1 = 1/2$, $D_1 = 12$, $H_1 = 8$ and $M_2 = 18$, $W_2 = 1$, $H_2 = 6$, $D_2 = ?$

$$\frac{10 * 12 * 8}{1/2} = \frac{18 * D_2 * 6}{1}$$

- $D_2 = 160/9$ days

Exercise

(Q.1) A man can do a piece of work in 5 days, but with the help of his son, he can do the same work in 3 days. In what time can the son do it alone? **[PI]**

(A) 13/2 (B) 7 (C) 15/2 (D) 8

(Q.2) A and B can do a piece of work in 72 days, B and C can do it in 120 days while A and C can do it in 90 days. In what time can A alone do it? **[PI]**

(A) 80 (B) 100 (C) 120 (D) 150

(Q.3) A can finish a work in 24 days, B in 9 days and C in 12 days. B and C start the work but are forced to leave after 3 days. The remaining work was done by A in? **[PI]**

(A) 5 (B) 6 (C) 10 (D) 11

(Q.4) X and Y can do a piece of work in 20 days and 12 days respectively. X started the work alone and then after 4 days Y joined him till the completion of the work. How long did the work last? **[PI]**

(A) 6 (B) 10 (C) 15 (D) 20

(Q.5) If 5 men or 9 women can do a piece of work in 19 days. In how many days will 3 men and 6 women do the same work? **[PI]**

(A) 10 (B) 12 (C) 13 (D) 15

(Q.6) A can complete a piece of work in 9 days and B in 12 days respectively. If they work for a day alternately. In how many days the work would be finished. If A begins the work? **[PI]**

(A) 31/4 (B) 41/4 (C) 43/4 (D) 11

(Q.7) A certain number of men can complete a job in 30 days. If there were 5 men more, it could be completed in 10 days less. How many men were in the beginning? **[PI]**

(A) 10 (B) 15 (C) 20 (D) 25

(Q.8) If 8 men, working 9 hours per day can build a wall 18 meters long, 2 meters wide and 12 meters high in 10 days. How many men will be required to build a wall 32 meters long, 3 meters wide and 9 meters high by working 6 hours a day in 8 days? **[PI]**

(A) 25 (B) 18 (C) 30 (D) 16

(Q.9) A can do a piece of work in 36 days, B in 54 days and C in 72 days. All the three began the work together, but A left before 8 days and B 12 days before the completion of the work. In how many days the work was finished from the day it stared? **[PI]**

(A) 24 (B) 28 (C) 36 (D) 48

(Q.10) 1 man or 2 women or 3 boys can do a work in 44 days. Then in how many days will 1 man, 1 woman and 1 boy do the work? **[PI]**

(A) 24 (B) 48 (C) 36 (D) 40

(Q.11) Two pipes can fill a tank in 20 and 24 minutes respectively and a waste pipe can empty 3 gallons per minute. All the three pipes working together can fill the tank in 15 minutes. The capacity of the tank is **[PI]**

(A) 180 (B) 150 (C) 120 (D) 60

(Q.12) Frog in the Well A frog is at the bottom of the well, which is 26 meters deep. Every day the frog jumps 6 meters upwards and falls 5 meters down. How many days will it take for the frog to reach the top? **[PI]**

(A) 21 Days (B) 23 Days (C) 24 Days (D) 26 Days

(Q.13) A and B can complete a piece of work in 8 days, B and C can do it in 12 days, C and A can do it in 8 days. A, B and C together can complete it in? **[PI]**

(A) 8 Days (B) 12 Days (C) 6 Days (D) 5 Days

(Q.14) A, B and C can do a piece of work in 20, 30 and 60 days respectively. In how many days can A do the work if he is assisted by B and C on every third day? **[PI]**

(A) 12 Dyas (B) 15 Days (C) 16 Days (D) 18 Days

(Q.15) 3 pumps, working 8 hours a day, can empty a tank in 2 days. How many hours a day must 4 pumps work to empty the tank in 1 day? **[PI]**

(A) 9 (B) 10 (C) 11 (D) 12

(Q.16) Frog in the well A frog is at the bottom of the well, which is 20 meters deep. Every day the frog jumps 5 meters upwards and falls 4 meters down. How many days will it take for the frog to reach the top? **[PI]**

(A) 16 Days (B) 17 Days (C) 20 Days (D) 15 Days

(Q.17) If 3 Men or 6 Women can complete task in 12 days, how long 6 Men and 8 women take to complete the same work? **[PI]**

(A) 5 Days (B) 4 Days (C) 3 Days (D) 6 Days

(Q.18) Anuj and Manoj can together paint their house in 30 days. After working for 20 days, Anuj has to go out and Manoj finishes the remaining work in the next 30 days. If Manoj had gone away after 20 days instead of Anuj, then Anuj would have completed the remaining work in? **[PI]**

(A) 18 (B) 12 (C) 15 (D) 20

(Q.19) A and B can complete a task in 12 days and 15 days respectively. They started the work together but B left 3 days before the completion. In how many days was the work completed? **[PI]**

(A) 8 (B) 6 (C) 9 (D) 10

(Q.20) From a group of 7 men and 6 women, a committee of 5 is to be formed such that at least 3 women are included. How many such committees are possible? **[PI]**

(A) 525 (B) 428 (C) 531 (D) 537

(Q.21) If it takes 5 machines 5 minutes to make 5 items, how long will 100 machines take to make 100 items? **[PI]**

(A) 5 minutes (B) 100 minutes (C) 50 minutes (D) 10 minutes

(Q.22) 39 persons can repair a road in 12 days, working 5 hours a day. In how many days will 30 persons, working 6 hours a day, complete the work? **[PI]**

(A) 10 (B) 13 (C) 14 (D) 15

Types of Series

1. ***Pure series:***
   - In this type series, the number itself obeys certain order so that the character of the series can be found out.
   - For e.g. Even number, Odd number, Prime number, Perfect square, Perfect cube, etc..

Ex. 2, 4, 6, 8, 10, ?

Sol. As per even series pattern, answer is 12.

Ex. 2, 3, 5, 7, 11, 13, ?

Sol. As per prime series pattern, answer is 17.

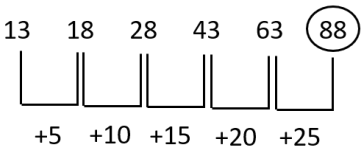
Ex. 6859, 5832, 4913, 4096, 3375, ?

Sol. As per perfect cube series pattern, answer is 2744.

2. ***Difference series:***
   - Under this category, the change in order for the differences between each consecutive number of the series is found out.

Ex. 13, 18, 28, 43, 63, ?

Sol.



3. ***Geometric series:***
   - Under this category, each successive number is obtained by multiplying (or dividing) the previous number with a fixed number.

Ex. 5, 35, 245, 1715

Sol. Here previous number is multiplied by 7, hence answer is 12005.

4. ***Mixed series:***
   - Under this category, the number obeying various orders of two or more different type of series are arranged alternately in a single number of series.

Ex. 7, 4, 5, 9, 20, ?

Sol. Here, second number is smaller than first one but from second number onwards, the series is in ascending order. So, there should be multiplication with fraction number or subtraction along with smaller multiplication.
   - 7 x 0.5 + 0.5 = 4, 4 x 1 + 1 = 5, 5 x 1.5 + 1.5 = 9, 9 x 2 + 2 = 20, 20 x 2.5 + 2.5 = 52.5
   - Hence, answer will be 52.5

5. ***Merged series:***
   - Under this category, usually two series are given with alternate position like one series is given for each odd place and second series is given for each even place.

Ex. 5, 25, 7, ? , 9, 19

Sol. One series is 5, 7, 9 (+2)
Second series is 25, ? , 19 (-3)
Hence, answer will be 25 - 3 = 22

Exercise

(Q.1) 100, 50, 52, 26, 28, ?

(A) 30 (B) 32 (C) 14 (D) 16

(Q.2) 25, 34, 30, 39, 35, ?

(A) 45 (B) 44 (C) 46 (D) 50

(Q.3) 32, ?, 92, 134, 184

(A) 55 (B) 38 (C) 45 (D) 58

(Q.4) 5, 12, 39, 160, 805, ?

(A) 4163 (B) 4153 (C) 4181 (D) 4836

(Q.5) 16, 8, 12, 30, ? , 472.5

(A) 100 (B) 105 (C) 205 (D) 300

(Q.6) 198, 194, 185, 169, ?

(A) 154 (B) 165 (C) 144 (D) 134

(Q.7) 5, 17, 37, 65, ?, 145

(A) 95 (B) 99 (C) 97 (D) 101

(Q.8) 812, 398, ?, 90, 40, 16

(A) 192 (B) 182 (C) 172 (D) 162

(Q.9) Find the odd man out: 198, 165, 148, 117, 104, 77

(A) 165 (B) 148 (C) 198 (D) 104

(Q.10) Find the odd man out: 7, 15, 42, 126, 231

(A) 42 (B) 231 (C) 126 (D) 15

(Q.11) 1, 4, 9, 16, _____, 25, 36. **[PI]**

(A) 9 (B) 23 (C) 25 (D) 36

(Q.12) Fill the blank in the middle of the series or end of the series: ZA5, Y4B, XC6, W3D, _____. **[PI]**

(A) E7V (B) V2E (C) VE5 (D) VE7

(Q.13) Look at this series: 7, 10, 8, 11, 9, 12, …. What number should come next? **[PI]**

(A) 7 (B) 10 (C) 12 (D) 13

(Q.14) Find the odd man out: 396, 462, 572, 427, 671, 264. **[PI]**

(A) 396 (B) 427 (C) 671 (D) 264

(Q.15) Which letter replaces the question mark: B, C, E, G,? **[PI]**

(A) K (B) M (C) H (D) None

(Q.16) Find the odd man out: 1, 5, 11, 17, 23, 29. **[PI]**

(A) 1 (B) 5 (C) 11 (D) 17

(Q.17) Look at this series: 22, 21, 23, 22, 24, 23, _____. What number should come next? [PI]

(A) 22 (B) 24 (C) 25 (D) 26

(Q.18) Look at this series: 53, 53, 40, 40, 27, 27, _____. What number should come next? [PI]

(A) 12 (B) 14 (C) 27 (D) 53

(Q.19) Look at this series: 21, 9, 21, 11, 21, 13, 21.... What number should come next? [PI]

(A) 14 (B) 15 (C) 21 (D) 23

(Q.20) Look at this series: 58, 52, 46, 40, 34, _____. What number should come next? [PI]

(A) 28 (B) 26 (C) 30 (D) 32

(Q.21) Look at this series: 36, 34, 30, 28, 24, _____. What number should come next? [PI]

(A) 20 (B) 22 (C) 23 (D) 26

(Q.22) 4, 3, 4, 9, 32, ... [PI]

(A) 125 (B) 135 (C) 145 (D) 155

(Q.23) Which term of the series 5, 8, 11,14, ... is 320? [PI]

(A) 104 (B) 105 (C) 106 (D) 64

(Q.24) In the series 5, 10, 20, 40,.... what will be the 10th term? [PI]

(A) 1280 (B) 2560 (C) 1820 (D) 2650

(Q.25) Find the next two terms of the series: 11 14 14 17 17 20 20. [PI]

(A) 21,21 (B) 23,23 (C) 25,25 (D) 21,23

(Q.26) Find the next two terms: 17, 32, 19, 29, 21, 26, 23. [PI]

(A) 25,25 (B) 23,29 (C) 23,27 (D) 23,25

(Q.27) Look at this series: J14, L16, P20, R22, ... What number should fill the blank? [PI]

(A) S24 (B) N18 (C) M18 (D) T24

(Q.28) Look at this series: V, VIII, XI, XIV, XX, ... What number should fill the blank? [PI]

(A) IX (B) XXIII (C) XV (D) XVII

(Q.29) Look at this series: 36, 34, 30, 28, 24, ... What number should come next? [PI]

(A) 20 (B) 22 (C) 23 (D) 26

(Q.30) Look at this series: 22, 21, 23, 22, 24, 23, ... What number should come next? [PI]

(A) 22 (B) 24 (C) 25 (D) 26

(Q.31) Look at this series: 53, 53, 40, 40, 27, 27.... What number should come next? [PI]

(A) 12 (B) 14 (C) 27 (D) 53

(Q.32) Look at this series: 21, 9, 21, 11, 21, 13, 21, ... What number should come next? [PI]

(A) 14 (B) 15 (C) 21 (D) 23

(Q.33) Look at this series: 58, 52, 46, 40, 34, ... What number should come next? [PI]

(A) 26 (B) 28 (C) 30 (D) 32

(Q.34) Look at this series: 14,28,20,40,32,64, ... What number should come next? [PI]

(A) 52 (B) 56 (C) 96 (D) 128

(Q.35) Look at this series: 2,6,18,54, ... What number should come next? [PI]

(A) 108 (B) 148 (C) 162 (D) 216

(Q.36) ELFA, GLHA, ILJA, _____, MLNA. [PI]

(A) JLLA (B) KMLA (C) KLLB (D) KLLA

(Q.37) 720, 720, 360, ___, 30, 6. [PI]

(A) 90 (B) 120 (C) 240 (D) 180

(Q.38) 3, 5, 5, 19, 7, 41, 9, __ . [PI]

(A) 71 (B) 49 (C) 81 (D) 63

(Q.39) 53, 53, 40, 40, 27, 27, __. [PI]

(A) 40 (B) 13 (C) 27 (D) 14

(Q.40) 21, 9, 21, 11, 21, 13, 21, __. [PI]

(A) 14 (B) 15 (C) 17 (D) 19

(Q.41) What are the next two letters in this series: A E F H I K L M ? ? [PI]

(A) P Q (B) O P (C) N T (D) O S

(Q.42) 25, 24, 22, 19, 15, ? [PI]

(A) 10 (B) 12 (C) 9 (D) 7

(Q.43) In the following question, one term in the number series is wrong. Find the wrong term. 3, 8, 15, 24, 35, 49, 63. [PI]

(A) 15 (B) 49 (C) 35 (D) 63

(Q.44) Complete the series 3, 7, 23, 95, ? [PI]

(A) 47 (B) 56 (C) 63 (D) 71

(Q.45) What is the next term in the sequence:1, 1, 2, 3, 5, 8, 13, ? . [PI]

(A) 18 (B) 21 (C) 19 (D) 26

(Q.46) In this pattern: 1, 4, 13, 40, 121, ?, what is the relationship? [PI]

(A) Each term is 3 times the previous + 1 (B) $3^n + 3^{n-1} + \ldots + 3^1 + 3^0$ (C) Powers of 3 minus 2 (D) Fibonacci sequence with different starting points

(Q.47) If 1=5, 2=25, 3=325, 4=4325, then 5=? [PI]

(A) 54325 (B) 54235 (C) 52345 (D) None of these

(Q.48) What is the next term in the sequence:1, 1, 2, 3, 5, 8, 13, ? [PI]

(A) 18 (B) 21 (C) 19 (D) 26

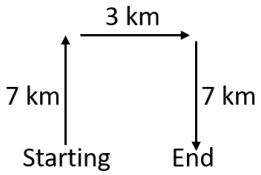
(Q.49) Look at this series: 120, 99, 80, 63, 48, ? What number should come next? [PI]

(A) 35 (B) 38 (C) 39 (D) 40

Introduction

Ex. Shyam travels 7 km North, then he turns to his right and walks 3 km. He again turns to his right and moves 7 km forward. Now in which direction is he from his starting point?

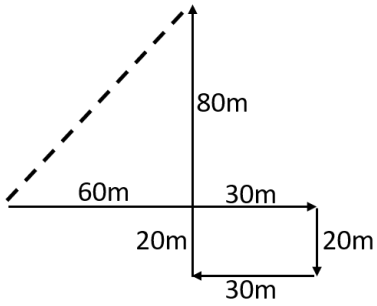
Sol.



- At the end he is East direction from his starting point.

Ex. A child is looking for his father. He went 90 meters in the east before turning to his right. He went 20 meters before turning to his right again to look for his father at his uncle's place 30 meters from this point. His father was not there. From there he went 100 meters to his north before meeting his father in a street. Now how far he is from his starting point?

Sol.



- Required distance = $\sqrt{80^2 + 60^2}$ = $\sqrt{6400 + 3600}$ = 100 m

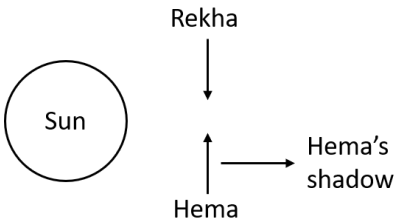
Ex. A man is facing North. He turns to 180° in clockwise direction and another 45° in the same direction and then 270° in the anticlockwise direction. In which direction is he facing now?

Sol. Let's take clockwise angle as + and anticlockwise angle as –

- 180° + 45° - 270° = - 45°
- Now, - 45° from North direction will be North-West direction.

Ex. In an evening Rekha and Hema were talking to each other face to face. If Hema's shadow was exactly to the right of Hema, which direction was Rekha facing?

Sol. In an evening, Sun is in West side.



- So, Rekha was facing South direction.

Exercise

(Q.1) Rohit start travelling toward North after going 20m. Then he takes a right turn and goes 30m then he takes a right turn again and goes 45m. Then he takes a left turn and goes 20m. Finally, he takes a left turn and goes 15m. In which direction is he from the starting point?

(A) South-East (B) North-East (C) North-West (D) South

(Q.2) Mr. Kamal goes 17 km in East then he takes a right turn and goes 52 km, then he takes a left turn and goes 7 km, again he takes a left turn and goes 21 km. Finally, he takes a left turn and goes 24 km. how far and in which direction is he from the starting point?

(A) 31km South (B) 32km East (C) 20km West (D) 7km East

(Q.3) Neha starts travelling from her office, initially she goes 20m, then she takes a left turn and goes 30m, again she takes a left turn and goes 40m. Finally, she takes a turn towards East and goes 10m. How far is she from the starting point?

(A) North (B) East (C) South (D) Can't be determined

(Q.4) One morning Amit and Sumit were talking to each other face to face. Amit's shadow was exactly to the right of Sumit. In which direction Sumit was facing?

(A) South (B) North (C) East (D) West

(Q.5) A man is facing west direction. He turns 180 in anticlockwise direction and another 45 in same direction and finally he moves in clockwise direction and moves 270. In which direction is he facing now?

(A) South-West (B) North-West (C) South (D) North

(Q.6) In morning Roshan covers 20 meters then he turns to his right and covers 20 meters. Again, he turns right and covers 5 meters. If at that time his shadow was towards his right then in which direction, he started his journey?

(A) North-East (B) North (C) South (D) East

(Q.7-9) Each of the following questions is based on the following information:

(1) Six flats on a floor in three are facing North and three are facing South direction. These flats are allotted to P, Q, R, S, T and X.
(2) Q gets a North facing flat and is not next to S.
(3) S and X get diagonally opposite flats.
(4) R is just right to X, gets a South facing flat and T gets North facing flat.

(Q.7) If the flats of P and T are interchanged, then who's flat will be next to that of X?

(A) P (B) Q (C) R (D) T

(Q.8) Which of the following combinations get South facing flat?

(A) QTS (B) XPT (C) XRP (D) PTX

(Q.9) The flat of which of the other pair than SX is diagonally opposite to each other.

(A) QP (B) QR (C) PT (D) TS

(Q.10) If South-East becomes North, North-East becomes West and so on. What will West become?

(A) South-East (B) North-East (C) North-West (D) South

(Q.11) In a 3D coordinate system, if you move 3 units north, 4 units east, and 12 units up from the origin, what is your distance from the origin? **[PI]**

(A) 19 (B) 12 (C) 13 (D) 25

(Q.12) A developer walks 10m North, then 10m East, then 10m South. Where is he from his original position? **[PI]**

(A) 10 m, East (B) 20 m East (C) 10 m, West (D) 20 m, West

Introduction

- To solve these problems, the basic knowledge of linear equations is essential.
- There may be three situations: Age some years ago, Present age and Age some years hence.

Ex. The ratio of present age of A & B is 2 : 3. The present age of A is 20 years. Find the age of B after 5 years.

Sol. The ratio of present age of A and B is given by $A^0 : B^0 = 2 : 3$, $A^0 = 20$ years.

- $2 \longrightarrow 20$, then $3 \longrightarrow 30$. So, present age of B is 30 years.
- $B^{+5} = 35$ years.

Ex. The present ratio of age of A & B is 3 : 5. If the sum of present age of A & B is 48 years. Find the ages of A & B before 5 years.

Sol. The ratio of present age of A and B is given by $A^0 : B^0 = 3 : 5$, so $(A+B)^0 = 8$.

- $8 \longrightarrow 48$, then $3 \longrightarrow 18$ and $5 \longrightarrow 30$. So, present ages of A & B are 18 and 30 years respectively.
- The ages of A & B before 5 years are 13 years and 25 years respectively.

Ex. The present ratio of ages of A & B is 6 : 7. After 5 years this ratio will be changed into 7 : 8. Find the present age of A & B.

Sol. The ratio of present age of A & B is given by $A^0 : B^0 = 6 : 7$, so $(A+B)^0 = 13$.

- The ratio of ages after 5 years is given by $A^{+5} : B^{+5} = 7 : 8$, so $(A+B)^{+5} = 15$.
- Actual difference between sum of present ages and sum of ages after 5 years should be 10 years.
- Therefore, $2 \longrightarrow 10$ then $6 \longrightarrow 30$ and $7 \longrightarrow 35$. So, the present ages of A & B are 30 years and 35 years respectively.

Ex. If Sita's mother was 4 times as old as Sita 10 years ago. After ten years, mother will be twice as old as Sita. How old is mother & Sita at present?

Sol. Let Sita's present age is X and Sita's mother's present age is Y.

- So, as per given data: $Y-10 = 4(X-10)$ & $Y+10 = 2(X+10)$
  $Y-10 = 4X-40$ & $Y+10 = 2X+20$
  $4X-Y = 30$ ……(1) & $2X-Y = -10$ ……(2)
- By solving these two equations, we will get $X = 20$ & $Y = 50$.
- Hence, Sita's present age is 20 years and Sita's mother's present age is 50 years.

Ex. Neeraj is as younger to Gopal as he is older to Deepak. If the sum of the ages of Gopal & Deepak is 58 years. What is Neeraj's age?

Sol. Gopal' age(G) – Neeraj's age(N) = Neeraj's age(N) – Deepak's age(D)

- G+D = 2N
- But we have G+D = 58 years, So 2N = 58
- Hence, Neeraj's age is 29 years.

Exercise

(Q.1) The ratio of ages of A & B before 5 years was 2:3. If the sum of ages of A & B at present is 45 years. Find the present ages of A & B.

(A) 19, 26 (B) 14, 21 (C) 24, 31 (D) 21, 14

(Q.2) The ratio of ages of A & B at present is 6:5 & the sum of their ages is 54 years after 5 years. What will be the ratio of their ages after 8 years?

(A) 7:8 (B) 4:5 (C) 8:7 (D) 14:11

(Q.3) Jayesh is as younger to Amit as Jayesh is older to Prashant. If the sum of the ages of Amit & Prashant is 48 years. What is the age of Jayesh?

(A) 20 (B) 24 (C) 28 (D) 25

(Q.4) The ratio of sum and the difference of the present age of Vivek and Arvind is 2:1. Four years ago, this ratio was 3:2. What will be the ratio of their ages after 12 years.

(A) 9:5 (B) 5:9 (C) 8:3 (D) 4:5

(Q.5) The ratio of ages of Ravi 5 years ago and that of Rohan's age after 5 years is 2:3. The ratio of their present age is 3:4. What is the age of Ravi before 3 years and that of Rohan's age after 7 years?

(A) 39:49 (B) 75:10 (C) 72:107 (D) 3:5

(Q.6) The ratio of present age of A and B is 11:13 and sum of their ages is 72 years. Then find the present age of C whose age is twice that of A?

(A) 56 (B) 12 (C) 78 (D) 66

(Q.7) The ratio of present age of Manoj and Waseem is 3:11. Waseem is 12 years younger than Rehana, 7 years hence Rehana will be 85 years old. What will be the present age of Manoj's father who us 25 years older than Manoj?

(A) 42 (B) 43 (C) 44 (D) 45

(Q.8) Ratio of present ages of A and B is 2:3. B's age after 16 years will be twice the age of A after 4 years. Find the difference between present ages of A and B.

(A) 9 (B) 11 (C) 2 (D) 8

(Q.9) The sum of the ages of a father and his son is 4 times the age of the son. If the average age of father and son is 28 years. What is the son's age?

(A) 14 (B) 16 (C) 12 (D) 10

(Q.10) After 5 years, the age of a father will be thrice the age of his son, whereas five years ago, he was 7 times as old as his son was. What are their present ages?

(A) 20, 30 (B) 10, 40 (C) 20, 40 (D) 10, 20

(Q.11) The present ages of three persons in proportions 4:7:9. Eight years ago, the sum of their ages was 56. Find their present ages in years. **[PI]**

(A) 8,14,18 (B) 8,20,28 (C) 16,28,36 (D) 20,35,45

(Q.12) The sum of the ages of 5 children born at the intervals of 3 years each is 50 years. What is the age of the youngest child? **[PI]**

(A) 4 (B) 8 (C) 10 (D) 12

(Q.13) Ten years ago, P was half of Q's age. If the ratio of their present ages is 3:4, what will be the total of their present ages? **[PI]**

(A) 45 (B) 40 (C) 35 (D) 30

(Q.14) A mother is twice as old as her son. If 20 years ago, the age of the mother was 10 times the age of the son, what is the present age of the mother? **[PI]**

(A) 45 (B) 40 (C) 35 (D) 30

(Q.15) The sum of the ages of 5 children born at the intervals of 3 years each is 50 years. What is the age of the youngest child? **[PI]**

(A) 4 (B) 8 (C) 10 (D) 12

(Q.16) If S were twice as old as he is, he would be 40 years older than J. J is 10 Years younger than S. How old is S? **[PI]**

(A) 20 (B) 30 (C) 40 (D) 50

(Q.17) If the ratio of ages of A and B is 3:4 and the sum of their ages is 56. What will be b's age after 4 years? **[PI]**

(A) 30 (B) 32 (C) 34 (D) 28

(Q.18) A man spent 1/3rd of his life as a bachelor. After ten years of marriage, his son was born. At this year's wedding anniversary, the father's age is double that of his son. What is the father's age at this year's anniversary? **[PI]**

(A) 55 (B) 60 (C) 65 (D) 54

Permutation

- The different arrangements of a given number or things by taking some or all at a time, are called permutation.
- For example, all permutations made with the letters of 'abc', taking two at a time are: ab, bc, ac, ca, bc, cb and all permutations made with the letters of 'abc', taking all at a time are: abc, acb, bac, bca, cab, cba.

Number of Permutation

- Number of all permutations of n things, taken r at a time, is given by:

$$^{n}P_{r} = \frac{n!}{(n-r)!}$$

Ex. In how many different ways can the letters of the word ROSE be arranged?

Sol. n = 4, r = 4

$$^{4}P_{4} = \frac{4!}{(4-4)!} = \frac{4!}{0!} = \frac{4*3*2*1}{1} = 24$$

Ex. In how many different ways can the letters of the word TOFFEE be arranged?

Sol. Here TOFFEE word has 6 letters but with repetition of F and E two times, therefore arrangement is

$$\frac{6!}{2!\,2!} = \frac{6*5*4*3*2*1}{2*2} = 180$$

Ex. In how many different ways can the letters of the word ELEPHANT be arranged so that vowels always occur together?

Sol. Vowels = E, E and A

They can be arranged in = $\frac{3!}{2!}$, now take these three vowels as one group, so we have total 6 letters

$$\frac{6!\,3!}{2!} = \frac{6*5*4*3*2*1 * 3*2*1}{2*1} = 2160$$

Combination

- Combination means selection, committee, group
- Each of the different groups or selections which can be formed by taking some or all of a number of objects called a combination.
- For example, if we want to select two out of three persons A, B, and C, then possible selections are: AB, BC and CA. (Note here that AB and BA represent the same selection)

Number of Combination

- The number of combinations of n things, taken r at a time is:

$$^{n}C_{r} = \frac{n!}{r!\,(n-r)!}$$

- $^{n}C_{n} = 1$, $^{n}C_{0} = 1$ and $^{n}C_{1} = n$

Ex. There are 15 persons in a group. They shake hand with each other. Find the difference number of hand shake.

Sol. At a time two persons can hand shake with each other, so total hand shake is

$$^{15}C_2 = \frac{15!}{2!\,(15-2)!} = \frac{15!}{2!\,13!} = \frac{15*14*13!}{2*1*13!} = 105$$

Ex. A committee of 5 members is to be formed out of 4 men and 5 women.

(i) In how many ways can a committee consisting of at least 1 woman be formed?
(ii) In how many ways can a committee consisting of 3 men and 2 women be formed?

Sol. (i) Committee has at least 1 woman, so there are different cases like 4 men + 1 woman, 3 men + 2 women, 2 men + 3 women, 1 men + 4 women and only 5 women.
Therefore, $^4C_4 \times {}^5C_1 + {}^4C_3 \times {}^5C_2 + {}^4C_2 \times {}^5C_3 + {}^4C_1 \times {}^5C_4 + {}^4C_0 \times {}^5C_5$
$= 1 \times 5 + 4 \times 10 + 6 \times 10 + 4 \times 5 + 1 \times 1$
$= 5 + 40 + 60 + 20 + 1 = 126$

(ii) Committee has 3 men and 2 women
Therefore, $^4C_3 \times {}^5C_2 = 4 \times 10 = 40$

Exercise

(Q.1) Find the number of different words that can be formed from the word 'SUCCESS'.

(A) 360 (B) 480 (C) 420 (D) 5040

(Q.2) In how many different ways can the letters of the word 'LEADING' be arranged in such a way that the vowels always come together?

(A) 360 (B) 720 (C) 650 (D) 4800

(Q.3) In how many ways the letters of the word 'WOMEN' can be rearranged so that when constant occupies odd places, vowel comes on even places?

(A) 10 (B) 11 (C) 12 (D) 13

(Q.4) Find the total number of employee code that can be formed by using two alphabets followed by 2 numbers and the letters should be distinct.

(A) 61000 (B) 54000 (C) 42000 (D) 63000

(Q.5) There are 5 professors, 8 lecturers and 7 teachers. Six people to be selected to make an interview board. In how many ways can they selected if it contains equal number of professors, lecturers and teachers?

(A) 5880 (B) 4770 (C) 3450 (D) 2180

(Q.6) A committee of 5 persons is to be formed from 6 men and 4 women. In how a=many ways can this be done when at most 2 women are included?

(A) 124 (B) 132 (C) 186 (D) 174

(Q.7) The total number of committee of 7 people is to be formed from 9 boys and 6 girls such that the boys are in the majority and it has at least 1 girl.

(A) 4914 (B) 2072 (C) 2076 (D) 3426

(Q.8) How many words can be made of letters COURTESY which began with C and end with Y?

(A) 120 (B) 540 (C) 420 (D) 720

(Q.9) A box contains 2 white balls, 3 black balls and 4 red balls. In how many ways can 3 balls be drawn from the box, if at least one black ball is to be included in the draw?

(A) 32 (B) 48 (C) 64 (D) 96

(Q.10) In how many ways 3 mathematics books, 4 history books, 3 chemistry books and 2 biology books can be arranged on a shelf so that all books of the same subjects are together.

(A) 42427 (B) 41472 (C) 44725 (D) 45722

(Q.11) In how many ways can the letters of the word 'TEACHER' be arranged? **[PI]**

(A) 2520 (B) 5040 (C) 360 (D) 720

(Q.12) 17 people at a party shake hands once with everyone else in the room. How many handshakes took place? **[PI]**

(A) 136 (B) 272 (C) 120 (D) 240

(Q.13) Michael has 10 pairs of black socks, 8 pairs of white socks, 5 pairs of red socks and 7 pairs of green socks. Everything is mixed in a draw. As there is no light he was not able to identify the color of the socks. What are the max, no. of socks she has to pull out in order to make sure that he gets one pair? **[PI]**

(A) 5 Socks (B) 6 Socks (C) 7 Socks (D) 8 Socks

(Q.14) There are 15 people in a group. They shake hands with each other. Find the different number of handshakes. **[PI]**

(A) 105 (B) 110 (C) 135 (D) 90

(Q.15) In how many ways can the letter of the word 'LEADER' be arranged? **[PI]**

(A) 72 (B) 144 (C) 360 (D) 720

(Q.16) Michael has ten pairs of black socks, eight pairs of white socks and seven pairs of green socks. Everything is mixed in a draw. As there is no light, she was not able to identify the color of the socks. What is the maximum no. of socks she has to pull out in order to make sure that she gets one pair? **[PI]**

(A) 5 (B) 6 (C) 4 (D) 7

(Q.17) In how many ways can you arrange 3 people in a row for a photograph. **[PI]**

(A) 3 (B) 6 (C) 9 (D) 12

(Q.18) In a company of 100 employees, 60 know Java, 50 know Python, and 20 know both. How many employees know neither language? **[PI]**

(A) 5 (B) 10 (C) 15 (D) 20

(Q.19) In how many different ways can the letters of the word 'CORPORATION' be arranged so that the vowels always come together? **[PI]**

(A) 810 (B) 1440 (C) 2880 (D) 50400

Introduction

Random experiment

- A random experiment is an experiment or a process for which the outcomes can not be predicted with certainty.
- For example, the process of rolling an unbiased dice, tossing a fair coin, drawing a card from a pack of well shuffled card, picking up a ball of certain color from a bag containing balls of different colors, etc… are known as random experiment.

Sample space

- The set of total possible outcomes in a random experiment is known as sample space denoted by S.
- For rolling a dice S = {1, 2, 3, 4, 5, 6}
- For tossing a coin once, S = {H, T}
- If two coins are tossed, then S = {HH, HT, TH, TT}

Probability

- Probability of any event (E), P(E) = n(E)/n(S)
  where, n(E) = Total number of required outcomes
  n(S) = Total number of possible outcomes
  P(E) = Probability of event

Ex. Find the probability of head when single coin is tossed.

Sol. n(E) = 1 {H}
n(S) = 2 {H, T}
P(E) = 1/2

Ex. Find the probability of at most one head when two coins are tossed simultaneously.

Sol. n(E) = 3 {TT, TH, HT}
n(S) = 4 {HH, HT, TH, TT}
P(E) = 3/4

Ex. Find the probability of getting a multiple of 3 when one dice is thrown once.

Sol. n(E) = 2 {3, 6}
n(S) = 6 {1, 2, 3, 4, 5, 6}
P(E) = 2/6 = 1/3

Ex. Find the probability that sum is divisible by 4 when two dice are thrown simultaneously.

Sol. n(E) = 9 {1+3, 2+2, 2+6, 3+1, 3+5, 4+4, 5+3, 6+2, 6+6}
n(S) = 36
P(E) = 9/36 = 1/4

Ex. If from a pack of 52 cards, 1 card is drawn at random. Find the probability that the card is an Ace card.

Sol. n(E) = ${}^{4}C_{1} = 4$
n(S) = ${}^{52}C_{1} = 52$
P(E) = 4/52 = 1/13

Exercise

(Q.1) An urn contains 4 green, 5 blue, 2 red and 3 yellow marbles. If eight marbles are drawn at random, what is the probability that there are equal number of marbles of each color?

(A) 4/7 (B) 361/728 (C) 60/1001 (D) 1

(Q.2) Two dice are rolled, find the probability that the sum is less than 13.

(A) ½ (B) 1 (C) 0.75 (D) 1.25

(Q.3) A bag contains 10 white balls and 16 black balls. Two balls are drawn in succession without replacement. What is the probability that first is white and second is black?

(A) 16/65 (B) 18/145 (C) 17/29 (D) 35/134

(Q.4) A bag contains 6 black, 5 white and 8 yellow balls. If four balls are picked at random, what is the probability that all four of them are either black or any two out of the four are white?

(A) 3/1292 (B) 925/3876 (C) 359/1927 (D) 23/24

(Q.5) If from a pack of 52 playing cards, 1 card is drawn at random. What is the probability that it is either a king or queen?

(A) 5/13 (B) 8/13 (C) 1/13 (D) 2/13

(Q.6) Two dice are tossed. The probability that the total score is an odd number?

(A) 1/6 (B) 5/12 (C) 1/2 (D) 7/9

(For Q.7-10) Rahul and Saif appear for an interview for two vacancies. The probability of Rahul's selection is 1/3 and that of Saif's selection is 1/5.

(Q.7) Find the probability that both of them will be selected.

(A) 1/15 (B) 8/15 (C) 8/9 (D) 7/8

(Q.8) Find the probability that none of them is selected.

(A) 1/15 (B) 8/15 (C) 8/9 (D) 7/8

(Q.9) Find the probability that at least one of them is selected.

(A) 1/15 (B) 8/15 (C) 8/9 (D) 7/15

(Q.10) Find the probability that only one of them is selected.

(A) 1/15 (B) 2/5 (C) 8/9 (D) 7/8

(Q.11) In a box, there are 8 red, 7 blue and 6 green balls. One ball is picked up randomly. What is the probability that it is neither red nor green? **[PI]**

(A) 1/3 (B) 3/4 (C) 7/19 (D) 8/21

(Q.12) What is the probability of getting a sum 10 from three throws of a dice? **[PI]**

(A) 25/216 (B) 1/27 (C) 1/8 (D) 10/216

(Q.13) What is the probability of getting an even number when a dice is rolled? **[PI]**

(A) 1/2 (B) 1/3 (C) 1/4 (D) 1

(Q.14) If from a pack of 52 playing cards, 1 card is drawn at random. What is the probability that it is either a king or queen? **[PI]**

(A) 2/13 (B) 4/13 (C) 7/52 (D) 5/13

(Q.15) What is the probability of getting a sum 9 from two throws of a dice? **[PI]**

(A) 1/6 (B) 1/8 (C) 1/9 (D) 1/12

(Q.16) There are 5 ants sitting on the five corners of a pentagon. All ants randomly pick a direction and start moving along edge of the pentagon. What is the probability that any two ants collide? **[PI]**

(A) 7/8 (B) 3/4 (C) 1/16 (D) 15/16

(Q.17) A box contains 6 red, 5 green, and 3 blue balls. Two balls are drawn without replacement. What is the probability that both are red? **[PI]**

(A) 10/91 (B) 1/5 (C) 2/11 (D) 15/91

(Q.16) In a box, there are 8 red, 7 blue and 6 green balls. One ball is picked up randomly. What is the probability that it is neither red nor green? **[PI]**

(A) 1/3 (B) 3/4 (C) 7/19 (D) 8/21

Introduction

- Mensuration deals with the study of different geometrical shapes, their area and volume.
- It is based on the use of algebraic equations and geometric calculations to provide measurement data regarding the width, depth and volume of a given object or group of objects.
- There are two types of geometric shapes:
  1) Two – dimensions (2D) shape: It has area and perimeter.
  2) Three – dimensions (3D) shape: It has volume and surface area.

Two – dimensions (2D)

Shape	Nomenclature	Area	Perimeter
Square	Side = a	a^2	$4a$
Rectangle	Length = l Breadth = b	l x b	$2(l+b)$
Triangle	Sides = a, b, c $s = \frac{a+b+c}{2}$	$\sqrt{s(s-a)(s-b)(s-c)}$	$a+b+c$
Circle	Radius = r	πr^2	$2\pi r$ (circumference)
Semicircle	Radius = r	$½\ \pi r^2$	$\pi r + 2r$

Ex. At the rate of Rs.2 per sq m, cost of painting a rectangular floor is Rs. 5760. If the length of the floor is 80% more than its breadth, then what is the length of the floor?

Sol. Let the length and the breadth of the floor be l and b respectively.

- $l = b$ + 80% of $b = b + 0.8\ b = 1.8\ b$
- Area of the floor = Total cost / Rate per sq m = 5760/2 = 2880 sq m
- $l * b$ = 2880
- l x $\frac{l}{1.8}$ = 2880
- l^2 = 5184
- l = 72, Length of the floor is 72 m.

Ex. Total area of a circle and a square is equal to 6350 m^2. The radius of a circle is 35 m. What is the perimeter of square?

Sol. Area of circle + Area of square = 6350

- $\pi r^2 + a^2$ = 6350
- $\frac{22}{7}(35)^2 + a^2$ = 6350
- a^2 = 6350 – 3850
- a^2 = 2500
- a = 50 m
- Perimeter of square = $4a$ = 4 x 50 = 200 m

Three – dimensions (3D)

Shape	Nomenclature	Volume	Surface area
Cube	Side = a	a^3	$6a^2$
Cuboid	Length = l Breadth = b Height = h	l x b x h	$2(lb + bh + hl)$
Cylinder	Radius = r Height = h	$\pi r^2 h$	Total surface area = $2\pi rh$ Curved surface area = $2\pi r(h + r)$
Cone	Radius = r Height = h Slant height = l	$\frac{1}{3}\pi r^2 h$	Total surface area = $\pi r(l + r)$ Curved surface area = πrl
Sphere	Radius = r	$\frac{4}{3}\pi r^3$	$4\pi r^2$
Hemisphere	Radius = r	$\frac{2}{3}\pi r^3$	Total surface area = $3\pi r^2$ Curved surface area = $2\pi r^2$

Ex. The height of the cone is 24 cm and radius of cone is 7 cm. Find the volume.

Sol. Radius r = 7 cm and height h = 24 cm

- Volume of cone = $\frac{1}{3}\pi r^2 h$

  = $\frac{1}{3}$ x $\frac{22}{7}$ x 7 x 7 x 24

  = 1232 cm^3

Exercise

(Q.1) A 7 m wide path is to be made around a circular garden having a diameter of 7 m. What will be the area of the path in square meter?

(A) 298 (B) 256 (C) 308 (D) 365

(Q.2) What is the volume of a cylinder whose curved surface area is 1408 cm^2 and height is 16 cm?

(A) 7715 cm^3 (B) 9340 cm^3 (C) 8722 cm^3 (D) 9856 cm^3

(Q.3) A cylinder whose base of circumference is 6 m can roll at a rate of 3 rounds per second. How much distance will the cylinder cover in 9 seconds?

(A) 125 m (B) 162 m (C) 149 m (D) 173 m

(Q.4) The radii of two cylinders are in the ratio 3 : 2 and their curved surface areas are in the ratio 3 : 5. What is the ratio of their volumes?

(A) 8 : 11 (B) 5 : 9 (C) 7 : 4 (D) 9 : 10

(Q.5) If a square, circle and rectangle has same perimeter then which one of them has the maximum area?

(A) Square (B) Circle (C) Rectangle (D) All are same

(Q.6) A square park has a 2m wide cross road in middle of it. If the side of park is 100m then find the remaining area of the park.

(A) 9650 m^2 (B) 9596 m^2 (C) 9600 m^2 (D) 9604 m^2

(Q.7) If the radius of a square increase by 4 cm then the surface area increases by 704 cm^2. The radius of the sphere initially was?

(A) 5 (B) 4 (C) 6 (D) 8

(Q.8) A rectangular plot of grass is 50m long and 40m broad. From the center of each side a path of 3m wide goes across the center of the opposite side. Find the area of path.

(A) 270 (B) 280 (C) 251 (D) 261

(Q.9) If the length of a rectangular plot is increased by 20% and the breadth of the plot is reduced by 20%, the area of the plot is decreased by 8 m^2. What is the original area of the rectangular plot?

(A) 184 (B) 196 (C) 200 (D) 225

(Q.10) If the perimeter of a circle is increased by 30%. What will be the effect on the area of the circle?

(A) 48% increase (B) 52% increase (C) 50% decrease (D) 69% increase

(Q.11) A person walked diagonally across a square plot. Approximately, what was the percent saved by not walking along the edges? **[PI]**

(A) 35% (B) 30% (C) 20% (D) 25%

(Q.12) If the radius of a circle increased by 20%, then what is the % change in area? **[PI]**

(A) 44% (B) 25% (C) 40% (D) 20%

(Q.13) A rectangle has length 8m and width 6m. What is its diagonal length? **[PI]**

(A) 14 m (B) 10 M (C) 12 m (D) 48 m

(Q.14) A cube is painted on all faces, then cut into 64 smaller identical cubes. How many small cubes will have exactly one face painted? **[PI]**

(A) 8 (B) 16 (C) 24 (D) 32

General Rule

1. ***To express $x\%$ as a fraction:***
   - We know $x\% = \frac{x}{100}$
   - Thus, $20\% = \frac{20}{100} = \frac{1}{5}$ ..... (means 1 part out of 5 parts)

2. ***To express $\frac{x}{y}$ as a percentage:***
   - We know that $\frac{x}{y} = \left(\frac{x}{y}\ X\ 100\right)\%$
   - Thus, $\frac{1}{4} = \left(\frac{1}{4}\ X\ 100\right)\% = 25\%$

3. ***Remember it:***

$50\% = \frac{1}{2}$	$33.33\% = \frac{1}{3}$	$14.28\% = \frac{1}{7}$	$7.69\% = \frac{1}{13}$
$25\% = \frac{1}{4}$	$16.67\% = \frac{1}{6}$	$7.14\% = \frac{1}{14}$	$6.67\% = \frac{1}{15}$
$12.5\% = \frac{1}{8}$	$8.33\% = \frac{1}{12}$	$11.11\% = \frac{1}{9}$	$5.88\% = \frac{1}{17}$
$6.25\% = \frac{1}{16}$	$4.16\% = \frac{1}{24}$	$9.09\% = \frac{1}{11}$	$5.26\% = \frac{1}{19}$

Ex. If x is 80% of y, what percent of x is y?

Sol. $y = \frac{100X100}{80} X\ x$

Therefore, $y = 125\%\ of\ x$

4. ***If A is R% more than B, then B is less than A by $\left[\frac{R}{100+R}\ X\ 100\right]\%$,***

   If A is R% less than B, then B is more than A by $\left[\frac{R}{100-R}\ X\ 100\right]\%$

Ex. If the income of A is 40% less than that of B, how much percent B's income is more than that of A?

Sol. $more\% = \frac{40}{100-40}\ X\ 100 = \frac{40}{60}\ X\ 100 = 66.67\%$

5. ***Net percentage change:***
   - With two successive increment or decrement in percentage (i.e. x% & y%), the net percentage change is given by $x + y + \frac{xy}{100}$
   - But always remember one thing x and y always come with sign. Means if quantity is increased, we will take + sign and if it is decreased, we will take - sign.

Ex. Find the net percentage change of 40% increase, 25% decrease and 10% increase?

Sol. For the first two changes, $40 - 25 - \frac{40X25}{100} = 5$

For the last change along with the net of first two, $5 + 10 + \frac{5X10}{100} = 15.5$

So, final change will be increase of 15.5%

Exercise

(Q.1) If income of Ram is 20% more than Shyam's income. How much percent Shyam's income is less than that of Ram's income?

(A) 12% (B) 20% (C) $16\frac{2}{3}\%$ (D) 30%

(Q.2) When numerator of a fraction is increased by 40% and denominator increased by 80%, the resultant fraction becomes $\frac{14}{27}$. Find the original fraction?

(A) $\frac{2}{3}$ (B) $\frac{3}{4}$ (C) $\frac{1}{3}$ (D) $\frac{4}{3}$

(Q.3) A number wrongly divided by 5, instead of multiplied by 5. Find the % error in result.

(A) 92% (B) 93% (C) 97% (D) 96%

(Q.4) If the radius of a circle increased by 20%, then what is the % change in area?

(A) 44% (B) 54% (C) 42% (D) 46%

(Q.5) A person spent 40% of his income. If his income is increased by 20%, then his expenditure also increased by 50%. Find the % change in his saving?

(A) 30% (B) 20% (C) 40% (D) No change

(Q.6) Two numbers are less than the third number by 50% and 54% respectively. By how much percent is the second number less than the first number?

(A) 13% (B) 8% (C) 4% (D) 10%

(Q.7) Weight of two friends Ram and Shyam are in the ratio of 4 : 5. Ram's weight increases by 10 % and the total weight of Ram and Shyam together becomes 82.8 kg, with an increase of 15%. By what % did the weight of Shyam increases?

(A) 17% (B) 18% (C) 19% (D) 20%

(Q.8) In a test, minimum passing percentage for girls and boys is 30% and 45% respectively. A boy scored 280 marks and failed by 80 marks. How much more marks did a girl require to pass in the test if she scored 108 marks?

(A) 132 (B) 140 (C) 160 (D) 112

(Q.9) Chhaya bought a watch costing Rs. 1404 including sales tax at 8%. She asked the shopkeeper to reduce the price of the watch so that she can save the amount equal to the tax. The reduction of the price of the watch is?

(A) Rs. 108 (B) Rs. 104 (C) Rs. 112 (D) Rs. 120

(Q.10) Two candidates A and B participate in election. 80% of the total people, cast the votes, in which A received 55% of the total votes casted. Difference between votes received by A and B is 1344. What is the number of registered votes?

(A) 14200 (B) 18200 (C) 16500 (D) 16800

(Q.11) In a certain store, the profit is 320% of the cost. If the cost increases by 25% but the selling price remains constant, approximately what percentage of the selling price is the profit? **[PI]**

(A) 30% (B) 100% (C) 70% (D) 250%

(Q.12) A shopkeeper sold an article for Rs. 2500. If the cost price of the article is 2000, find the profit percent. **[PI]**

(A) 20% (B) 25% (C) 30% (D) 10%

(Q.13) 40% of 280 = _____. **[PI]**

(A) 122 (B) 112 (C) 110 (D) 160

(Q.14) A number is increased by 20% and then decreased by 20%. The net change is? **[PI]**

(A) 8% Decrease (B) 4% increase (C) 4% decrease (D) 0%

(Q.15) A man marks an article 60% above its cost price. He allows two successive discounts of 25% and 20% and still gains 80. What is the cost price of the article? **[PI]**

(A) 500 (B) 1000 (C) 2000 (D) 1500

Ratio

- In a ratio, two quantities are compared, so the quantities must be of the same kind.
- The ratio of two quantities determines how many times one quantity is contained by the other.

Type - 1: Derive ratio of three quantities from given sets of two ratios with one common quantity.

- If A : B = x : y and B : C = p : q then,
  1) A : C = x p : y q
  2) A : B : C = (x : y) p : y q = x p : y p : y q

Ex. If A : B = 3 : 4 and B : C = 8 : 9 then A : B : C is?

Sol. A : B = (3 : 4) 2 = 6 : 8
B : C = 8 : 9
A : B : C = 6 : 8 : 9

Type - 2: Derive ratio of four quantities from given sets of three ratios with one common quantity in group of two ratios.

- If A : B = x : y, B : C = p : q and C : D = m : n, then
  1) A : D = x p m : y q n
  2) A : B : C : D = xpm : ypm : yqm : yqn

Ex. If A : B = 2 : 3, B : C = 4 : 5 and D : C = 7 : 6 then A : B : C : D is?

Sol. A : B = 2 : 3
B : C = 4 : 5
C : D = 6 : 7
A : B : C : D = 2 x 4 x 6 : 3 x 4 x 6 : 3 x 5 x 6 : 3 x 5 x 7 = 16 : 24 : 30 : 35

Proportion

- A proportion is an expression which states that two ratios are equal.
- E.g. $\frac{3}{12} = \frac{1}{4}$ is a proportion, it can also be expressed as 3 : 12 = 1 : 4 or 3 : 12 :: 1 : 4

Type - 1: Third proportion.

- If a : b :: b : c, then c is called the 3rd proportional to a and b.
- c will be calculated as follow: a : b :: b : c or a : b = b : c
- a c = b b, therefore c = $\frac{b^2}{a}$

Ex. Find the third proportion of 0.36 & 0.48.

Sol. Third proportion c = $\frac{b^2}{a} = \frac{0.48 * 0.48}{0.36}$ = 0.64

Type - 2: Fourth proportion.

- If a : b :: c : d, then d is called the 4th proportional to a, b and c.
- d will be calculated as follow: a : b :: c : d or a : b = c : d
- a d = b c, therefore d = $\frac{bc}{a}$

Ex. Find the fourth proportion of 3, 6, & 12.

Sol. Third proportion d = $\frac{bc}{a} = \frac{6 * 12}{3}$ = 24

Type - 3: Mean proportion.

- If a and b are two numbers then mean proportion is $\sqrt{ab}$

Ex. Find the mean proportion between 9 and 16.

Sol. Mean proportion = $\sqrt{9 * 16} = \sqrt{144}$ = 12.

Ex. A bag contains 50 P, 25 P and 10 P coins in the ratio of 5 : 9 : 4 amounting to Rs. 206. Find the number of coins of 50 paise.

Sol. Let the number of 50 P, 25 P and 10 P coins be 5x, 9x and 4x respectively.
First, we have to convert rupees to paise to equalize currency. So, Rs. 206 = 20600 paise.

50 * 5x + 25 * 9x + 10 * 4x = 20600
250x + 225 x + 40 x = 20600
515x = 20600
x = 40

Therefore, number of 50 paise coins = 5 * 40 = 200

Ex. A mixture contains alcohol and water in the ratio of 4 : 3. If 5 liters of water is added to the mixture, the ratio becomes 4 : 5. Find the quantities of alcohol in the given mixture.

Sol. Let the quantity of alcohol and water be 4x liters and 3x liters respectively.

4x / (3x + 5) = 4 / 5
20x = 12x + 20
8x = 20
x = 2.5

Quantity of alcohol = 4 * 2.5 = 10 liters.

Exercise

(Q.1) A watermelon is cut into two pieces in the ratio of 3 : 5 by weight. The bigger of the two is further cut in the ratio of 5 : 7 by weight. Find the ratio of weight of each of the three pieces.

(A) 3 : 5 : 7 (B) 15 : 25 : 56 (C) 36 : 25 : 35 (D) 3 : 2 : 3

(Q.2) Divide Rs. 3600 among P, Q, R in the ratio 7 : 2 : 9. Find the amount received by Q.

(A) 1000 (B) 400 (C) 500 (D) 1800

(Q.3) The monthly incomes of two persons are in the ratio of 2 : 3 and their monthly expenses are in the ratio 5 : 9. If each of them saves Rs. 600 per month then the total of their monthly incomes is?

(A) 4500 (B) 4000 (C) 4800 (D) 6000

(Q.4) The length and breadth of a rectangle are raised in the ratio 3 : 4 and 4 : 5 corresponding. What is the ratio of the previous area to the new area?

(A) 5 : 3 (B) 3 : 5 (C) 5 : 8 (D) 8 : 5

(Q.5) Rs. 6800 is divided among three person A, B & C in such a way that A received 2/5 of total shared B & C together. Find the share of B.

(A) 1800 (B) 2000 (C) 2200 (D) Can't find

(Q.6) The incomes of A & B are in the ratio of 5 : 7& their expenditures are in the ratio of 3 : 5. If each saves Rs.1500. Find the income of B.

(A) 4250 (B) 5250 (C) 3550 (D) 4050

(Q.7) Rs.525 is divided among 4 men, 5 women and 6 boys such that share of a man, a woman and a boy may be in the ratio of 9 : 8 : 4. What is the share of a women?

(A) 36 (B) 38 (C) 40 (D) 42

(Q.8) A vessel contains 60 liters of mixture of milk and water in the ratio of 7 : 3 respectively. 8 liters of mixture is replaced by 12 liters of milk. What is the ratio of milk and water in the resulting mixture?

(A) 67 : 121 (B) 121 : 39 (C) 164 : 123 (D) 165 : 121

(Q.9) The price of sugar is increased by 20%. If the expenditure is not allowed to increase, the ratio between the reduction in consumption and the original consumption is?

(A) 1 : 3 (B) 1 : 4 (C) 1 : 6 (D) 4 : 1

(Q.10) The largest and smallest angles of a triangle are in the ratio 3 : 1. The second largest angle of the triangle is equal to 44°. What is the value of 150% of the largest angle of the triangle?

(A) 149 (B) 129 (C) 153 (D) 155

(Q.11) Two numbers have a ratio of 2:3. If their sum is 20, then what is the sum of the squares of two numbers? **[PI]**

(A) 324 (B) 200 (C) 196 (D) None of these

(Q.12) If A:B = 2:3 and B:C = 4:5, what is A:C? **[PI]**

(A) 4:10 (B) 2:5 (C) 8:15 (D) 3:4

(Q.13) A and B invest in a business in the ratio 3:2. If 5% of the total profit goes to charity and A's share is Rs. 855, the total profit is: **[PI]**

(A) 1425 (B) 1500 (C) 1537.50 (D) 1576

General Rule

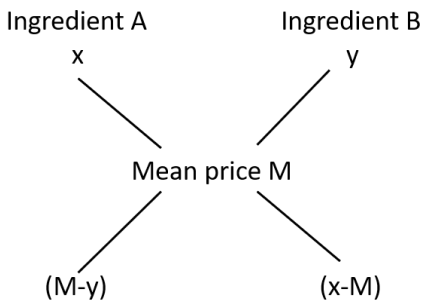
1. ***Alligation:***
   - It enables us to find the ration in which two or more ingredients at the given price must be mixed to produce a mixture of a desired price.

2. ***Mean price:***
   - The cost price of a unit quantity of the mixture is called the mean price.

3. ***Basic formula:***
   - If two ingredients are A and B are mixed of price x and y respectively, then the price of resultant mixture is M mean price.
   - The ratio in which ingredients are mixed is given by rule of alligation.

$$R = \frac{M - y}{x - M}$$

   - The above formula can be represented as

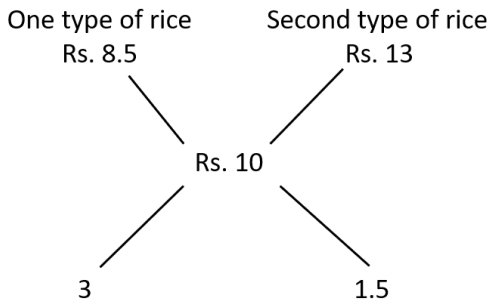


   - Thus, the required ratio is

$$R = \frac{M - y}{x - M} = \frac{y - M}{M - x}$$

Ex. In what ratio must rice costing Rs. 8.50 per kg be mixed with rice costing Rs. 13 per kg so that the mixture be worth Rs. 10 per kg?

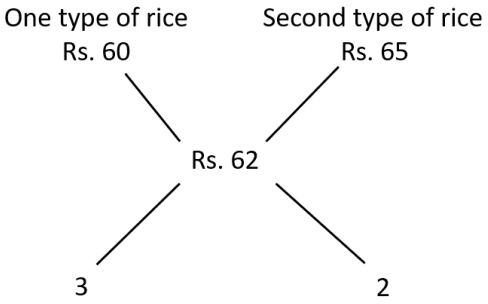
Sol. Here, by cross connection formula, 10 - 8.5 = 1.5 & 13 -10 = 3 as shown below.



Required ratio = 3 : 1.5 = 2 : 1

Ex. In what ratio must a grocer mix two varieties of sugar costing Rs.60 per kg and Rs.65 per kg, so that on selling the mixture at Rs.68.20 per kg, he may gain 10%?

Sol. Cost price of 1 kg of mixture = 68.20 x $\frac{100}{110}$ = 62



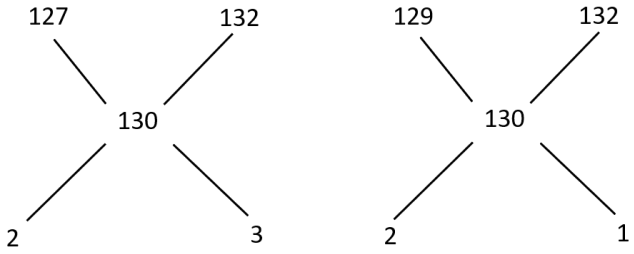
Required ratio = 3 : 2

4. Mixture of more than two elements:

- These questions may seem to be a little tricky at first, but similar concept is applied repeatedly in order to calculate final ratio of ingredients when mixture contains more than two ingredients.
  1. Take two ingredients in such a way that first ingredient is lower than the mean value and the other one is higher than mean value.
  2. Calculate the ratio of ingredients.
  3. Repeat for all possible pairs.
  4. Final ratio is the ratio obtained from step-2 (if an ingredient is common in the ratios, add value for this particular ingredient)

Ex. Three types of Rice of Rs. 1.27, Rs. 1.29 and Rs. 1.32 per kg are mixed together to be sold at Rs. 1.30 per kg. In what ratio should this rice be mixed?

Sol. Here, we can convert fractional number into integer by simply multiply with 100 to each and every rupees and by cross connection formula, we will get ratio as shown below

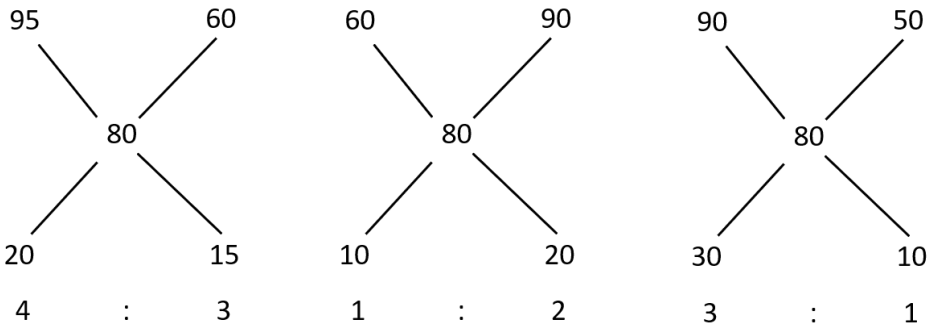


Hence, final ratio is 2 : 2 : 3+1 = 2 : 2 : 4 = 1 : 1 : 2

5. Mixture containing 4 ingredients:

Ex. How much a shop owner mixture 4 types of rice worth Rs. 95, Rs. 60, Rs. 90 & Rs. 50 per kg so that he can make the mixture of these rice worth Rs. 80 per kg.

Sol. Here, we have to mix two mixtures at a time with total three different types of mixing in which one quantity should remain common in every beside mixture, like 95 & 60, 60 & 90 and 90 & 50 as shown in below.



Hence, final ratio is 4 : 3+1 : 2+3 : 1 = 4 : 4 : 5 : 1

Exercise

(Q.1) What quantity of tea costing Rs. 8 per kg be mixed with 15 kg of tea costing Rs. 20 per kg so that mixture be worth Rs. 12 per kg?

(A) 15 kg (B) 20 kg (C) 25 kg (D) 30 kg

(Q.2) Two alloys are both made up of copper and tin. The ratio of copper and tin in the first alloy is 1 : 3 and in the second alloy is 2 : 5. In what ratio should the two alloys be mixed to obtain a new alloy in which ratio of tin and copper be 8 : 3?

(A) 7 : 3 (B) 5 : 7 (C) 4 : 7 (D) 7 : 4

(Q.3) 1050 rupees are distributed among 1400 males and females such that each male got Rs. 1 and each female got 50 paisa. Find the number of females among the group.

(A) 600 (B) 700 (C) 800 (D) 900

(Q.4) A cricketer whose bowling average is 24.85 runs per wicket takes 5 wickets for 52 runs and there by decrease his average by 0.85. The number of wickets taken by him before the last match was?

(A) 64 (B) 72 (C) 80 (D) 96

(Q.5) How many kg of salt at 42 paisa per kg must be a man mixed with 25 kg of salt at 24 paisa per kg, so that he gains 25% on outlay on selling the mixture at 40 paisa per kg?

(A) 15 kg (B) 20 kg (C) 25 kg (D) 30 kg

(Q.6) A man has 90 pens. He sells some of these at a profit of 15% and rest at 9% profit. On the whole transaction he gets a profit of 11%. How many pens did he sell at 9% profit?

(A) 60 (B) 50 (C) 40 (D) 70

(Q.7) Weight of two friends Ram and Shyam are in the ratio of 4 : 5. Ram's weight increases by 10 % and the total weight of Ram and Shyam together becomes 82.8 kg, with an increase of 15%. By what % did the weight of Shyam increases?

(A) 17% (B) 18% (C) 19% (D) 20%

(Q.8) 12 liters of water is drawn out from a container full of water and replaced by milk. Again 12 liters of mixture is drawn and the container is again filled with milk. The ratio of final quantity of water to milk in the container is 11 : 25. How much water did the container hold?

(A) 60 liters (B) 65 liters (C) 72 liters (D) 39 liters

(Q.9) A man travelled a distance of 90 kms in 9 hours, partly on foot at 8 km/hr and partly on bicycle at 17 km/hr. Find the distance travelled on foot.

(A) 50 km (B) 56 km (C) 60 km (D) 55 km

(Q.10) In what ratio must a person mix three kind of tea each of which has a price of 70, 89 & 120 rupees per kg in such a way that the mixture cost 100 Rs./kg.

(A) 2:3:4 (B) 2:1:2 (C) 5:2:2 (D) 2:2:5

(Q.11) A mixture contains alcohol and water in the ratio of 4:3. If 5 liters of water is added to the mixture, the ratio becomes 4: 5. Find the quantities of alcohol in the given mixture. **[PI]**

(A) 10 litters (B) 15 litters (C) 12 litters (D) 9 litters

(Q.12) In what ratio must rice costing Rs. 8.50 per kg be mixed with rice costing Rs. 13 per kg so that the mixture be worth Rs. 10 per kg? **[PI]**

(A) 2:1 (B) 1:2 (C) 1:1 (D) 2:3

(Q.13) Sugars that cost 9 per kg and 7 per kg are mixed. Starting with 27 kg of 7 per kg sugar, how many kilograms of 9 per kg sugar must be added to provide a 10% gain when sold at 9.24 per kg? **[PI]**

(A) 65 (B) 49 (C) 63 (D) 72

(Q.14) A dishonest grocer professes to sell milk at cost. In truth, the grocer mixes it with water to gain 25% against the original cost. What percentage of the mixture is water? **[PI]**

(A) 20% (B) 25% (C) 30% (D) 15%

(Q.15) In what ratio must a grocer mix two varieties of pulses costing Rs. 15 and Rs. 20 per kg respectively so as to get a mixture worth Rs. 16.50 kg? **[PI]**

(A) 7:3 (B) 5:7 (C) 7:5 (D) 3:7

General Rule for Calendar

- Month number

Month	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Normal	1	4	4	0	2	5	0	3	6	1	4	6
Leap	0	3	4	0	2	5	0	3	6	1	4	6

- Century number

Century	Number
1600	6
1700	4
1800	2
1900	0
2000	6
2100	4

- Day number

Day	Sun	Mon	Tue	Wed	Thu	Fri	Sat
Number	1	2	3	4	5	6	0

- Formula = (Day + Month + Year + Century + Leap years in between) % 7

Ex. What was the day of the week on 28th May, 2006?

Sol. Here, Day = 28, Month is May = 2, Year = 6, Century is 2000 = 6, Leap year between 2000 & 2006 = 1

Formula = (28 + 2 + 6 + 6 + 1) % 7

= 43 % 7

= 1

Therefore, the day of the week on 28th May, 2006 was Sunday.

General Rule for Clock

To solve problems regarding hand movement, remember these fixed ratios:

- Total Clock Face: 360°.
- Subsequent Divisions (Hours): Each hour mark is 30° apart (360/12).
- Minute Hand: Moves 6° per minute (360/60).
- Hour Hand: Moves 0.5° per minute (30/60).
- Second Hand: Moves 6° per second (360/60).

Formula to find angle between hour and minute hands:

$$\text{Angle} = \left|30H - \frac{11}{2}M\right|$$

Where H = Hours and M = Minutes.

Exercise

(Q.1) It was Sunday on Jan 1, 2006. What was the day of the week Jan 1, 2012?

(A) Sunday (B) Saturday (C) Friday (D) Wednesday

(Q.2) What was the day of the week on 17th June, 1998?

(A) Monday (B) Tuesday (C) Wednesday (D) Thursday

(Q.3) What will be the day of the week 15th August, 2010?

(A) Sunday (B) Monday (C) Tuesday (D) Friday

(Q.4) Today is Monday. After 61 days, it will be:

(A) Wednesday (B) Saturday (C) Tuesday (D) Thursday

(Q.5) If 6th March, 2005 is Monday, what was the day of the week on 6th March, 2004?

(A) Sunday (B) Saturday (C) Tuesday (D) Wednesday

(Q.6) On 8th Feb, 2005 it was Tuesday. What was the day of the week on 8th Feb, 2004?

(A) Tuesday (B) Monday (C) Sunday (D) Wednesday

(Q.7) If it was a Sunday on Jan 1, 2006. The day of the week on Jan 1, 2010 is _____ **[PI]**

(A) Sunday (B) Saturday (C) Friday (D) Wednesday

(Q.8) How many days are there in x weeks x days? **[PI]**

(A) 7x2 (B) 8x (C) 14x (D) 7

(Q.9) On 5th December, 1993, Raju celebrated his birthday on Sunday. What will be the day of his birthday in 1997? **[PI]**

(A) Wednesday (B) Thursday (C) Friday (D) Saturday

(Q.10) If today is Thursday, after 730 days which will be the day of the week? **[PI]**

(A) Thursday (B) Friday (C) Saturday (D) Monday

(Q.11) If January 1st, 2024 was a Monday, what day of the week will January 1st, 2025be? **[PI]**

(A) Monday (B) Wednesday (C) Tuesday (D) Thursday

(Q.12) In the time in which the second hand covers 3960 degrees, how many degrees does the hour hand move? **[PI]**

(A) 11 degrees (B) 5.5 degrees (C) 3/4 degrees (D) 55 Degrees

(Q.13) By how many degrees does the minute hand move at the same time, in which the hour hand moves by 18 degrees? **[PI]**

(A) 168 (B) 216 (C) 196 (D) 276

(Q.14) What is the angle between the two hands of a clock when the time shown by the Clock is 5.30 P.M.? **[PI]**

(A) 0 (B) 5 (C) 3 (D) 15

(Q.15) Angle between subsequent divisions in clock (360 degrees)? **[PI]**

(A) 30 (B) 90 (C) 120 (D) 180

(Q.16) The angle between the minute hand and the hour hand of a clock when the time is 2:10 is: **[PI]**

(A) 10 (B) 0 (C) 5 (D) 20

1. Seating Arrangements (Linear & Circular)

Seating arrangements test your ability to translate relative positions into a fixed sequence.

Linear Arrangement (Rows/Benches)

- **Directionality:** If the row faces **North**, "right" matches your right hand; if they face **South**, the directions are reversed.
- **End-Point Clues:** Always look for statements like "at one of the ends" or "4th to the right of G" to establish a framework.
- **Negation:** Clues like "D is not sitting with E" help eliminate possible neighbors when multiple slots are open.

Circular Arrangement

- **Facing the Center:** If everyone faces the center, "Right" is **counter-clockwise** and "Left" is **clockwise**.
- **Facing Outside:** If they face away from the center, "Right" is **clockwise** and "Left" is **counter-clockwise**.
- **Diagonals:** In an even-numbered group (6 or 8), "diagonally opposite" means there are an equal number of people between the two subjects on both sides.

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2. Ranking and Ordering

Ranking problems involve calculating a person's position relative to the whole group.

- **The Total Formula:** To find the total number of people in a row:

  Total = (Position from Start + Position from End) - 1.

- **Finding the Opposite Rank:** To find a rank from the bottom:

  Bottom Rank = (Total - Top Rank) + 1.

- **Overlapping Queues:** If two people swap positions, the total remains constant. Use the new position of one person and their original position from the other side to find the total.

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3. Complex Logic Puzzles (Scheduling & Attributes)

These require a **Grid/Matrix approach** to organize multiple variables like names, days, and subjects.

- **Fixed Constraints:** Fill in certainties first (e.g., "B teaches Chemistry on Tuesday").
- **Negative Constraints:** Mark cells with an "X" for things that *cannot* happen (e.g., "D does not teach Geography").
- **Commonality:** If two subjects are taught on the same day, and that day already has one subject assigned, they must both occupy that day's slots.

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4. Mathematical Logic & Riddles

These problems often rely on optimization or specific strategies rather than standard arithmetic.
The "Bridge and Torch" Problem

To find the shortest time for a group to cross a bridge with one light:

- **Overlap Slow Times:** Send the two slowest people across together so their times overlap.
- **Return with the Fastest:** Always use the person with the fastest speed to bring the torch back to the starting side.

Weighing and Probabilities

- **Divide and Conquer:** For weighing identical-looking items to find one "heavy" item, divide the group into three equal parts. Weighing two parts against each other tells you which of the three groups contains the outlier.
- **Worst-Case Scenario (Socks/Gloves):** To guarantee a pair when picking in the dark, you must assume you pick one of every color first. The very next pick will then *guarantee* a match.

Venn Diagram Analysis

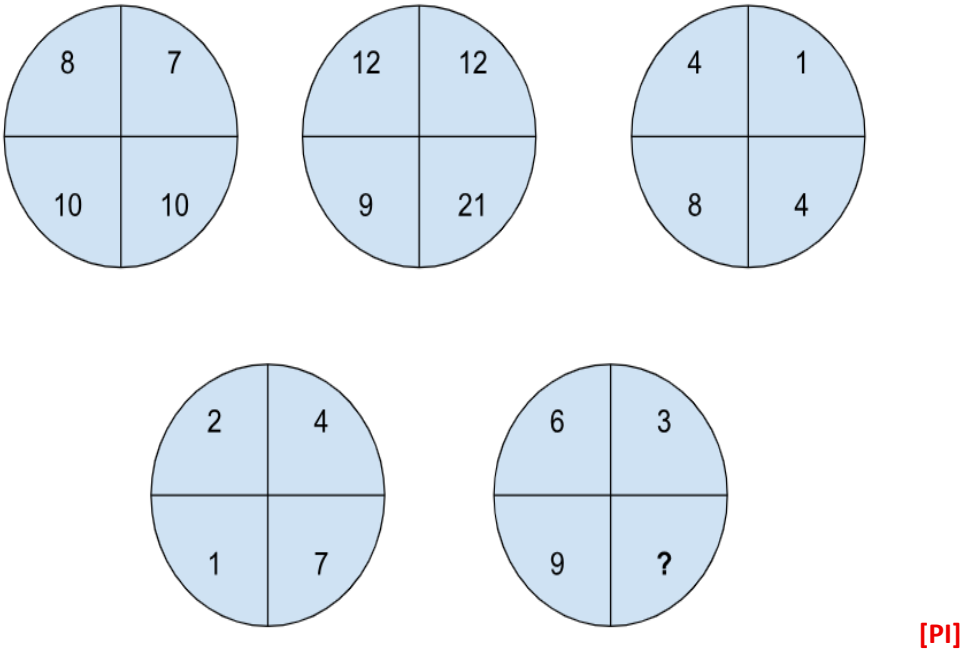
- **Intersections:** "Strong and Tall but not Army" requires finding the area where the "Strong" and "Tall" shapes overlap, but the "Army" shape does not.

Exercise

Q.1 A, B, C, D and E are sitting on a bench. A is sitting next to B, C is sitting next to D, D is not sitting with E who is on the left end of the bench. C is on the second position from the right. A is to the right of B and E. A and C are sitting together. Then the persons on either side of A are _____ **[PI]**

(A) B, D (B) B, C (C) E, D (D) C, E

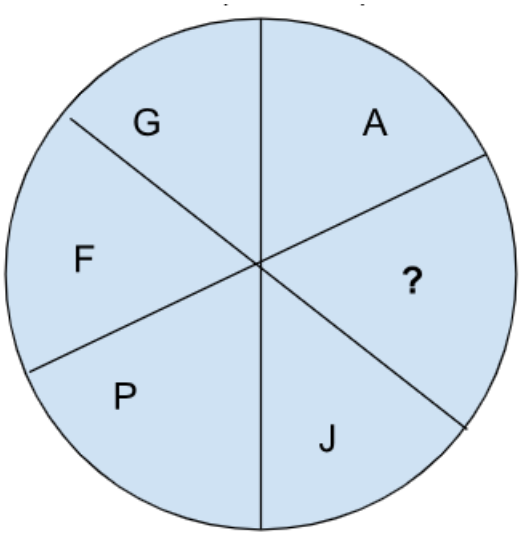
Q.2



[PI]

(A) 3 (B) 4 (C) 5 (D) 6

Q.3



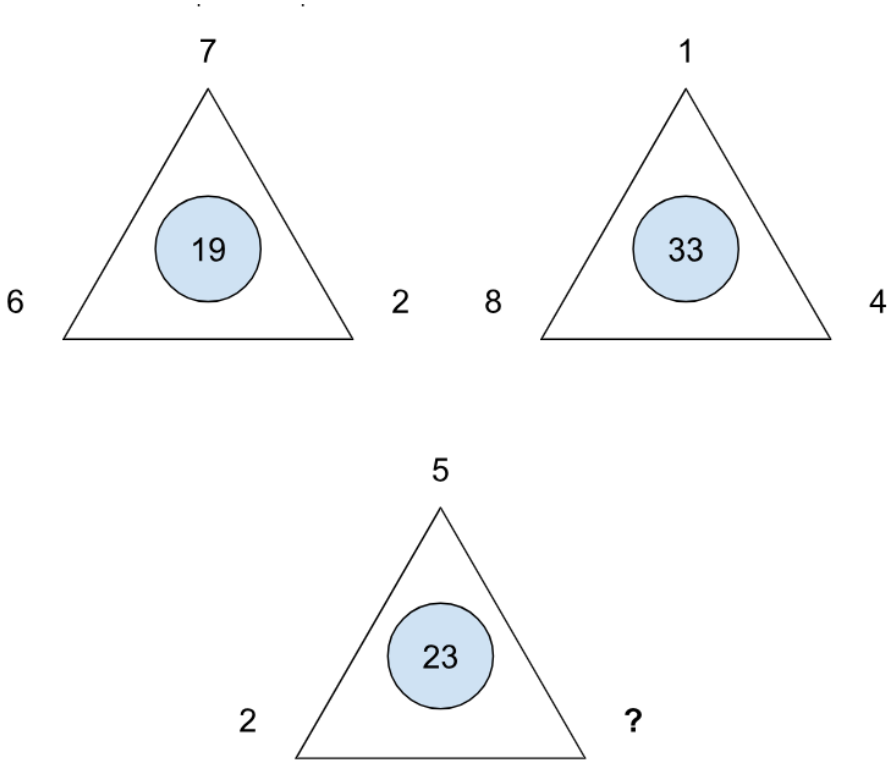
[PI]

(A) S (B) T (C) U (D) R

Q.4 There are 9 identical balls. In that one of the balls is overweight. There are only two chances to weigh in the weighing machine. How to find the overweight ball? Write steps only. [PI]

Q.5 A milkman carries a full 12 liters container of milk. He needs to deliver exactly 6 liters to a customer who only has 8-liter and 5-liter containers. How can he do this? Write steps only. [PI]

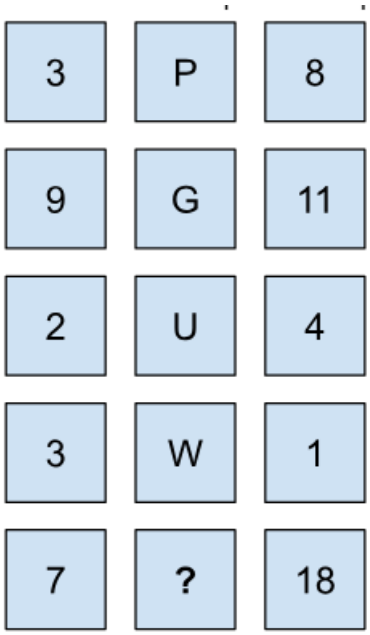
Q.6



[PI]

(A) 10 (B) 9 (C) 8 (D) 6

Q.7



[PI]

(A) R (B) S (C) T (D) U

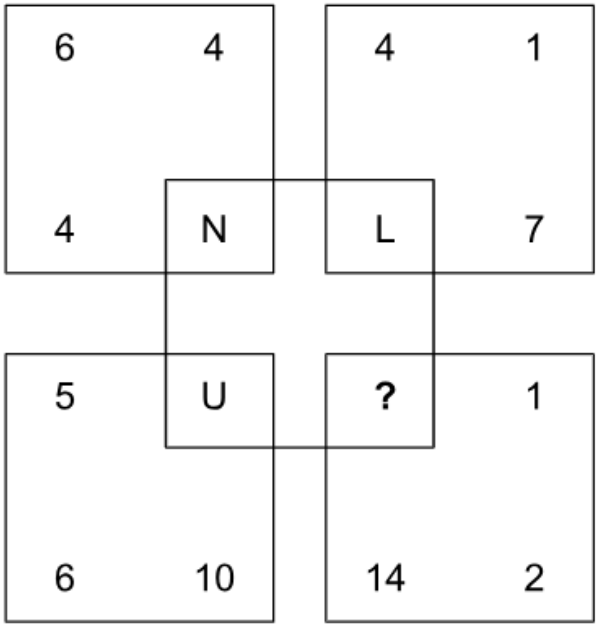
Q.8 Michael has ten pairs of black socks, eight pairs of white socks and seven pairs of green socks. Everything is mixed in a draw. As there is no light she was not able to identify the color of the socks. What is the maximum no. of socks she has to pull out in order to make sure that she gets one pair? [PI]

(A) 4 Socks (B) 5 Socks (C) 6 Socks (D) None of these

Q.9 Four people need to cross a rickety bridge at night. Unfortunately, they have only one torch and the bridge is too dangerous to cross without one. The bridge is only strong enough to support two people at a time. Not all people take the same time to cross the bridge. Times for each person: 1 min, 2 min, 7 min and 10 min. What is the shortest time needed for all four of them to cross the bridge? [PI]

(A) 23 mins (B) 24 mins (C) 25 mins (D) None of these

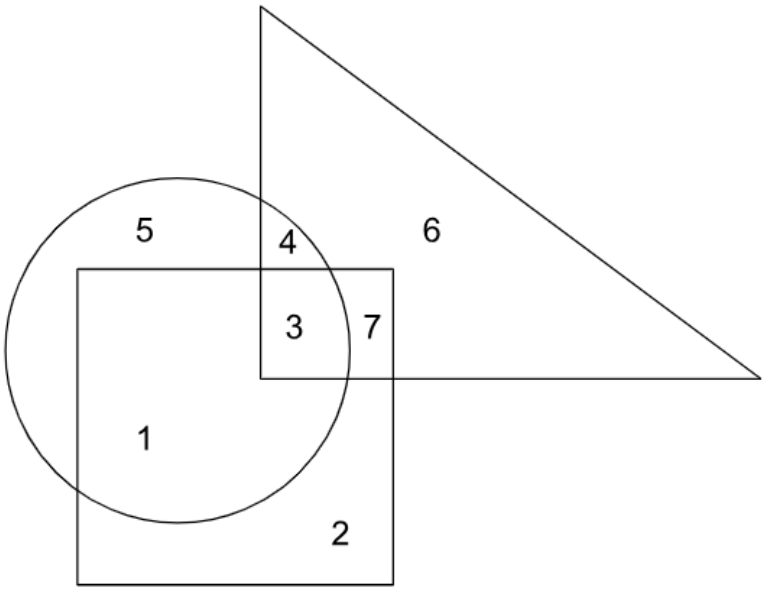
Q.10



[PI]

(A) R (B) S (C) Q (D) T

Q.11 Circle indicates 'strong', the square indicates 'tall' and the triangle 'army officers'. The strong army officers who are not tall are shown.



[PI]

(A) 4 (B) 3 (C) 5 (D) 6

Q.12 Michael has 10 pairs of black socks, 8 pairs of white socks, 5 pairs of red socks and 7 pairs of green socks. Everything is mixed in a draw. As there is no light he was not able to identify the color of the socks. What are the max, no. of socks she has to pull out in order to make sure that he gets one pair? **[PI]**

(A) 5 socks (B) 6 Socks (C) 7 socks (D) None of these

Q.13 John and his family members want to cross to the other side of the bridge at night. They have only one lamp which lasts for only 30 minutes. A maximum of only two persons can cross at one time. and they must have the lamp with them. Each person walks at a different speed, John walks at a speed of 1 min, his brother Jack walks at 3 min, his mother Julie walks at 6 min, his father Jeff walks at 8 min and his grandpa George walks at 12 min. A pair must walk together at the rate of the slower person. How can John's family cross the bridge? **[PI]**

(A) 26 minutes (B) 30 minutes (C) 25 minutes (D) None of these

Q.14-Q.15 Six men B, D, C, M, J and K are split in two groups of three each and are made to stand in two rows, such that a man in one row is exactly facing a man in the other row.
M is not at the ends of any row and is to the right of J, who is facing C. K is to the left of D. who is facing M. **[PI]**

Q.14 Which of the following groups of men are in the same row?

(A) BMD (B) MJK (C) BDC (D) None of these

Q.15 Who is to the immediate left of B?

(A) M (B) D (C) J (D) None of these

Q.16-Q.20 There are seven professors A, B, C, D, E, F and G teaching seven subjects History, Geography, Physics, Chemistry, Maths, Biology and English from Monday to Friday at Gaya College. Each professor teaches a different subject and not more than two subjects are taught on any one of the days. **[PI]**

1. Chemistry is taught by professor B on Tuesday.
2. Professor D teaches on Friday but neither Geography nor Physics.

3. Professor F teaches History but neither on Thursday nor on Friday.
4. Professor A teaches English on the day on which History is taught
5. Professor C teaches Maths on Monday.
6. Geography and Chemistry are taught on the same day.
7. Professor G teaches on Thursday.

Q.16 On which of the following days is English taught?

(A) Wednesday (B) Monday (C) Tuesday (D) None of these

Q.17 Which of the following subjects is taught by professor G?

(A) Biology (B) Geography (C) Physics (D) Chemistry

Q.18 On which of the following days is Geography taught?

(A) Monday (B) Tuesday (C) Wednesday (D) Thursday

Q.19 Which subject is taught on Friday?

(A) Physics (B) History (C) Geography (D) Biology

Q.20 Which of the following pairs of professors teaches on Tuesday?

(A) B and D (B) A and B (C) B and F (D) None of these

Q.21 – Q.24 In a class there are seven students (including boys and girls) A, B, C, D, E, F and G. They sit on three benches I, II and Ill. Such that at least two students on each bench and at least one giri on each bench. C who is a girl student, does not sit with A, E and D. F the boy student sits with only B. A sit on the bench I with his best friends. G sits on the bench III. E is the brother of C. **[PI]**

Q.21 How many girls are there out of these 7 students?

(A) 3 (B) 3 or 4 (C) 4 (D) Data inadequate

Q.22 Which of the following is the group of girls?

(A) BAC (B) BFC (C) BCD (D) CDF

Q.23 Who sits with C?

(A) B (B) D (C) G (D) E

Q.24 On which bench there are three students?

(A) Bench I (B) Bench II (C) Bench III (D) Bench I or Bench II

Q.25 – Q.28 Amit, Bharat, Chandan, Dinesh, Eeshwar and Ferguson are cousins. None of them are of the same age, but all of them have birthdays on the same date. The youngest of them is 17 years old and Eeshwar, who is the eldest, is 22 years old. Ferguson is somewhere between Bharat and Dinesh in age. Amit is elder to Bharat and Chandan is older than Dinesh. **[PI]**

Q.25 Which of the following is not possible?

(A) Dinesh is 20 years old (B) Ferguson is 18 years old (C) Ferguson is 19 years old (D) Ferguson is 20 years old

Q.26 If Bharat is 17 years old, then which of the following could be the ages of Dinesh and Chandan respectively?

(A) 18 and 19 (B) 19 and 21 (C) 18 and 20 (D) None of these

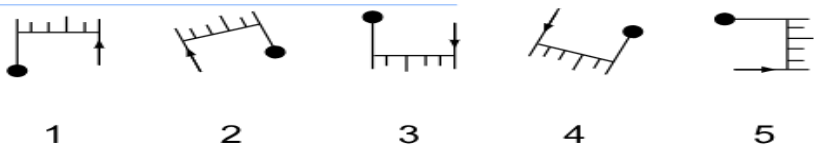
Q.27 If two of the cousins are between Chandan and Ferguson in age, then which of the following must be true?

(A) Amit is between Ferguson and Dinesh
(B) Bharat is 17 years old
(C) Bharat is younger than Dinesh
(D) Ferguson is 18 years old

Q.28 If Chandan is 19 years old, which of the following must be true?

(A) Amit is 20 years and Dinesh is 21 years
(B) Bharat is 18 years and Amit is 20 years
(C) Bharat is 20 years and Amit is 21 years
(D) Dinesh is 17 years and Bharat is 21 years

Q.29 Out of the five figures, four are similar in a certain manner. However, one figure is not like the other four. Choose the figure which is different from the rest. **[PI]**



(A) 1 (B) 2 (C) 3 (D) 4

Q.31 – Q.35 K, L, M, N, P, Q, R, S, U and W are the only ten members in a department. There is a proposal to form a team from within the members of the department subject to the following conditions: **[PI]**

1. A team must include exactly one among P, R, and S.
2. A team must include either M or Q, but not both
3. If a team includes K, then it must also include L, and vice versa.
4. If a team includes one among S, U, and W, then it must also include the other two.
5. Land N cannot be members of the same team
6. L and U cannot be members of the same team
7. The size of a team is defined as the number of members in the team.

Q.31 Who cannot be a member of a team of size 3?

(A) L (B) M (C) N (D) P

Q.32 Who can be a member of a team of size 5?

(A) K (B) L (C) M (D) P

Q.33 What would be the size of the largest possible team?

(A) 8 (B) 7 (C) 6 (D) 5

Q.34 What could be the size of a team that includes K?

(A) 2 or 3 (B) Only 2 (C) Only 4 (D) None of these

Q.35 n how many ways can a team be constituted so that the team includes N?

(A) 3 (B) 4 (C) 5 (D) 6

Q.36 – Q.38 P, Q, R, S, T, U, V and W are sitting round the circle and are facing the centre:

P is second to the right of T who is the neighbour of R and V.

S is not the neighbour of P.

V is the neighbour of U.

Q is not between S and W. W is not between U and S. **[PI]**

Q.36 Which two of the following are not neighbours?

(A) RV (B) UV (C) RP (D) QW

Q.37 One is immediate right to the V?

(A) P (B) U (C) R (D) T

Q.38 Which of the following is correct?

(A) P is to the immediate right of Q (B) R is between U and V (C) Q is to the immediate left of W (D) U is between W and S

Q.39 – Q.43 Four people A, B, C and D went to four different places which is Kerala, Delhi, Goa and Gujarat by different transports as follows- Train, Car, Aeroplane and Bus. The person who travelled to Delhi did not travel by Bus. C travelled by Aeroplane and went to Goa. A went to Kerala by Car and D travel by Bus. **[PI]**

Q.39 The person who travels by Train?

(A) is going to Delhi (B) is going to Kerala (C) is C (D) is D

Q.40 Who among the following person is going to Kerala?

(A) A (B) B (C) C (D) D

Q.41 Who among the following person is going to Delhi?

(A) A (B) B (C) C (D) D

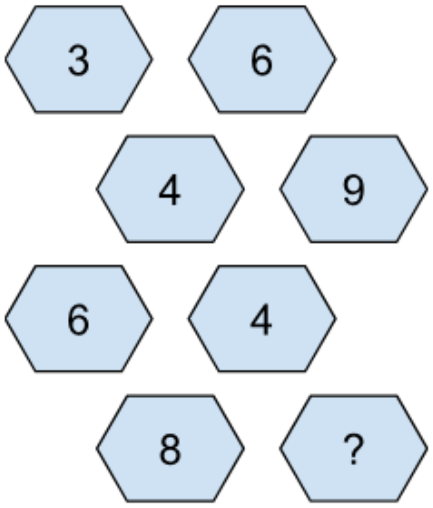
Q.42 Which person is going to Goa?

(A) A (B) B (C) C (D) D

Q.43 The person who travels by Car is?

(A) B (B) C (C) Going to kerala (D) Going to Goa

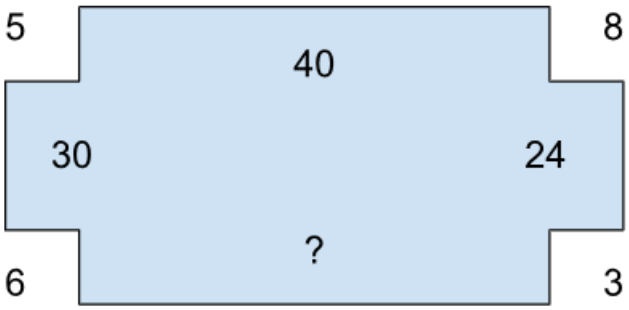
Q.44 Which number should be in the place of "?".



[PI]

(A) 1 (B) 2 (C) 3 (D) 4

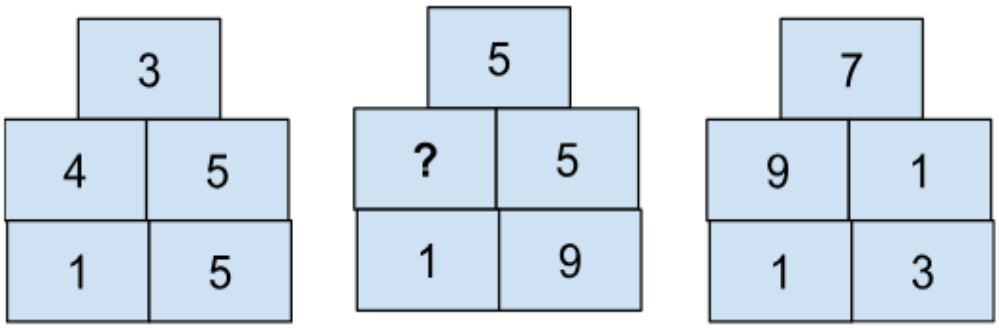
Q.45 Find the number in the missing place?



[PI]

(A) 18 (B) 19 (C) 22 (D) 38

Q.46



[PI]

(A) 4 (B) 6 (C) 8 (D) 9

Q.47 – Q.49 A, B, C, D, E, F and G are sitting in a row facing North

F is to the immediate right of E.

E is 4th to the right of G.

Person who is third to the left of D is at one of the ends.

C is the neighbour of B and D. [PI]

Q.47 How many people are there exactly between B and D?

(A) 0 (B) 1 (C) 2 (D) 3

Q.48 How many people are there to the left of C?

(A) 2 (B) 3 (C) 4 (D) 5

Q.49 What is the position of D?

(A) Between B and F (B) Between E and A (C) Between G and C (D) Between C and E

Q.50 Reshma and Praveena are ranked ninth and thirteenth from the top in a class of 57 students. What will be their respective ranks from the bottom of the class? [PI]

(A) 48,44 (B) 49,45 (C) 45,49 (D) None of these

Q.51 – Q.54 Six players K, M, G, D, L and T are sitting in a circular table facing the center. M is to the left of K. D is between G and T. L is between G and M. and M is diagonally opposite to D [PI]

Q.51 How many people are seated between L and K if counted clockwise?

(A) 1 (B) 2 (C) 3 (D) 4

Q.52 How many people are seated between L and D if counted clockwise?

(A) 3 (B) 2 (C) 1 (D) 4

Q.53 Who sits diagonally opposite to L?

(A) T (B) D (C) K (D) M

Q.54 Who sits to the right of L?

(A) G (B) M (C) T (D) None of these

Q.55 In a class of 90 students, where girls are twice that of boys, Shridar ranked fourteenth from the top. If there are 10 girls ahead of Shridar, how many boys are after him in rank? **[PI]**

(A) 23 (B) 26 (C) 25 (D) 22

Q.56 A family went for a vacation. Unfortunately, it rained daily for 13 days when they were there. But whenever it rained in the mornings, they had clear afternoons and vice versa. In all, they enjoyed 11 mornings and 12 afternoons. How many days in all did they stay there? **[PI]**

(A) 25 (B) 18 (C) 21 (D) 22

Q.57 In a queue, Amrita is 10th from the front while Mukul is 25th from behind and Mamta is just in the middle of the two. If there are 50 persons in the queue. What position does Mamta occupy from the front? **[PI]**

(A) 15 (B) 16 (C) 17 (D) 18